

ORIGINAL ARTICLE

Harnessing the Hybrid Intelligence of Crowd and Artificial Intelligence in Group Decision Making for Uncertain Disaster Response

Ming Tang | Mengjia Li | Leilei Xie 

School of Economics and Management, Dalian University of Technology, Dalian, China

Correspondence: Leilei Xie (xieleilei0511@163.com)

Received: 10 July 2025 | **Revised:** 12 March 2026 | **Accepted:** 27 March 2026

Keywords: AI | crowdsourcing | group decision making | selective bias risk | uncertainty

ABSTRACT

Crowdsourcing is now widely used in disaster management. However, restricted access to social media of vulnerable people in poor areas will pose a risk of selective bias. This study presents a framework to harness the hybrid intelligence of crowd and artificial intelligence (AI) in group decision-making problems to improve the inclusiveness of disaster response decisions. First, crowdsourced data are utilized to extract immediate needs, while an AI algorithm predicts broader population requirements based on area vulnerability features. To improve compassion, the degree of human suffering is estimated by a deep learning method, so as to allocate relief resources to the most urgent victims. To manage the inherent uncertainties in decision-making, a stochastic optimization model is used. Next, a group decision-making method is proposed by incorporating the solutions of the crowd and the AI. A consensus-building approach through mining the maximum consensus sequences is introduced. The framework's efficacy is demonstrated through a real-world case study of the 2021 Henan flood. This study contributes to the field by offering a comprehensive model that enhances real-time disaster response through the combined strengths of crowd and AI intelligence in group decision making.

1 | Introduction

Crowdsourcing is an innovative tool that leverages collective intelligence to solve complex problems (Zhao and Zhu 2014). In disaster response, crowdsourcing enables real-time collection and processing of disaster-related data, providing support for rapid decision making (Alhaffar et al. 2023; Abbasi et al. 2024). In recent years, the emergence of crowdsourcing platforms has changed the traditional response mode. Crowdsourcing platforms leverage digital technologies to collect information and process tasks across crowds. Crowds can provide first-hand information by reporting their situation through digital media and can also take the initiative to participate in rescue operations, such as providing temporary shelters, sharing material information, and

participating in volunteer rescue services. Figure 1 is an example of a crowdsourcing post.

Although the use of crowdsourcing platform has increased in recent years, it faces challenges because of “selective bias” (Esparza et al. 2023). The digital divide due to unequal access to social media platforms will distort situational awareness generated during disasters and lead to an underestimation of some victims' needs (Dargin et al. 2021). Many rural areas face problems such as unstable network signals, insufficient bandwidth, and high access costs, which directly hinder residents from using crowdsourcing platforms (Graves et al. 2021). Vulnerable groups (including the elderly, children, women, people with disabilities, etc.) face challenges in accessing and using social



FIGURE 1 | A crowdsourcing post.

media, especially in the use of novel digital platforms. The digital divide is further widened by the lack of digital skills and the ability to use information and communication technologies by vulnerable groups (Morte-Nadal and Esteban-Navarro 2025). Moreover, under-education leads to a lack of necessary digital literacy to become proficient in operating complex digital platforms. As a result, these social and geographical biases exacerbate the vulnerability of rural residents to disasters and hinder the realization of the potential of crowdsourcing platforms.

Another issue lies that even with real-time crowdsourced data, humanitarian organizations still face substantial uncertainties when making their group decisions. Information gathered from crowds during disasters is often of low quality. Victims driven by feelings of tension and fear may exaggerate their circumstances and the quantity of supplies they need. Such exaggeration could lead to the misallocation of crucial resources and unsuccessful humanitarian operations. Disaster situations can also change at any time. Because of communication delays, the demands of affected people cannot be immediately adjusted. Hence, humanitarian organizations should address parameter uncertainty, namely, the uncertainty of estimations, in decision-making. Such uncertainty affects the subsequent alternative evaluation and consensus-building process in group decision making (GDM).

To address the aforementioned challenges, this study builds a framework to address disaster response decision-making problems based on crowdsourcing platform posts. First, we gather rescue estimates of trapped vulnerable people using crowdsourced and artificial intelligence (AI) approaches, respectively. Crowd estimates are determined by crowdsourcing and mining posts, whereas AI estimates are based on the geographic distribution of the crowd estimates and vulnerability features. Second, we establish a stochastic optimization model to obtain decision-making solutions of the crowd and the AI. The humanitarian literature commonly defines probability distributions as representing the uncertainty caused by parameters. Given a specific probability measure, a set of scenarios is generated to approximate the stochastic model. Finally, we introduce a consensus-building method to obtain consensus in GDM with a set of experts in related fields.

The contributions of this study are as follows.

1. We propose an innovative collaborative decision-making framework integrating the hybrid intelligence of crowd and AI to address selective bias. This framework integrates crowd and AI estimates of the number of vulnerable people in need of rescue. Crowds can transmit and update disaster information in a timely manner, which is crucial for

evaluating the needs of victims in ever-changing environments. AI algorithms can rapidly learn emergency-related patterns from large-scale data and are helpful in compensating for the rescue information loss caused by the geographic disparities of social media use. By combining crowd and AI, rather than replacing one with the other, objectives can be achieved that cannot be accomplished effectively with one intelligence source alone. Moreover, alleviating human suffering is a main goal of humanitarian operations (Holguín-Veras et al. 2012). Trapped individuals may endure significant mental and physical distress, including pain, cold, hunger, and fear, particularly if they lack essential goods and services for an extended period (Holguín-Veras et al. 2016). To gauge the extent of real-time suffering, we use a text classification technique. Next, crowd and AI estimates will be used as alternative inputs of the GDM model. Therefore, this framework aims to leverage crowd, AI, and group experts in a comprehensive manner.

2. We propose a method to generate a set of scenarios considering the degree of conflict between the crowd and AI estimates. The humanitarian literature commonly defines probability distributions to represent the uncertainties caused by parameters (e.g., the number of vulnerable people in need of rescue). To minimize the computational load, a clustering method is used to obtain representative scenarios. A stochastic optimization model is built to obtain the crowd and AI solutions.
3. We develop a method to achieve group consensus in GDM. In real-world emergency decision-making problems, organizational experts usually have the final rights to make decisions because of their human experience and authority. To do this, crowd and AI solutions are served as feasible alternatives of GDM. Thereafter, a consensus model based on mining maximum consensus sequences is introduced to improve the degree of consensus.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 obtains crowd and AI decisions based on a stochastic optimization model. Section 4 introduces a group decision-making model with crowd-AI solutions and a consensus-building approach. Section 5 summarizes the procedure of the proposed model. Section 6 provides a case study. Finally, Section 7 presents the concluding remarks.

2 | Literature Review

In this section, we review the applications of crowdsourced intelligence, AI, and GDM in disaster management in recent years.

2.1 | Crowdsourcing in Disaster Management

Crowdsourcing can be defined as outsourcing work traditionally done by designated personnel (typically organizational staff) to undefined public groups through open recruitment (Howe 2006). Crowdsourcing first shown great potential in the field of disaster management in the 2010 Haiti earthquake (Keim and Noji 2011). Thereafter, crowdsourcing has been increasingly popular in

disaster management because of its capacity to address complex response challenges and enable scalable task completion.

Kankanamge et al. (2019) investigated the role of volunteer crowdsourcing in disaster risk reduction. Tavra et al. (2021) integrated multi-source disaster data collected via Crowdfunder, social media platforms (Twitter/Facebook), and fire department communications with authoritative satellite imagery. This synthesis significantly enhanced situational awareness while optimizing rescue resource allocation and enabling rigorous fire dynamics analysis in forest fire response. Bahadori et al. (2022) employed the CrowdBIG system to collect multidimensional earthquake disaster data, including victim status and critical infrastructure conditions. They ensured data reliability through automated verification and integrated geolocation technology to generate comprehensive disaster maps. Nowadays, social media has transformed disaster response by providing real-time, first-hand information from affected individuals (Yoo et al. 2024). Social media platforms, such as Weibo and Twitter, have been instrumental in past disasters as they enable passive crowdsourcing (Lam et al. 2023), where content is extracted to inform supply and rescue requests. Loynes et al. (2022) developed a prototype portal to identify local and regional non-government organizations through harnessing the power of crowdsourcing from social media. Alhaffar et al. (2023) improved disaster response operations using community-driven crowdsourcing. This approach enabled rapid localization of disaster zones and resource planning, significantly enhancing the efficiency and accuracy of relief efforts.

In flood disaster scenarios, Alizadeh et al. (2022) used crowdsourcing platforms (e.g., Blupix) to collect user-generated street inundation imagery. They generated high-fidelity dynamic flood data for emergency response and optimized evacuation routes, thus reducing disaster impacts. Using passive Twitter data harvesting alongside active online questionnaire distribution, Moghadas et al. (2023) established a dual crowdsourcing mechanism. This framework enabled real-time capture of disaster dynamics and public perception while resolving chronic deficiencies in information timeliness and public participation within conventional disaster systems. In the face of the hurricane disaster, Cheng et al. (2024) recruited participants through the Amazon Mechanical Turk platform to assist AI in completing building damage assessments. Khajwal and Noshadravan (2024) improved crowdsourcing efficiency by developing Amazon MTurk micro-task questionnaires. These questionnaires documented building structural details and roof/wall damage metrics. Nielsen et al. (2024) provides a state-of-the-art of the research filed linking social media and crowdsourcing with disaster risk management based on a comprehensive review of 237 papers.

In existing research, crowdsourcing has been widely used in various disaster management scenarios, and has significantly improved the timeliness of information acquisition and the accuracy of disaster relief decision-making. However, a significant concern is the “selective bias” or “digital divide” where data contributions disproportionately represent certain demographic groups, potentially overlooking vulnerable populations (Dargin et al. 2021). This can lead to an imbalanced situational awareness and reinforce existing social inequities during disasters (Esparza

et al. 2023). Therefore, improving the fairness and inclusiveness of crowdsourcing remains a critical research frontier.

2.2 | AI in Disaster Management

AI, particularly machine learning (ML) and deep learning (DL), has become a transformative tool for processing complex disaster data and developing decision support systems across the entire disaster management lifecycle. In recent years, fruitful studies have outlined the applications, challenges, and future directions of AI in various phases of disaster risk management.

During the mitigation and preparedness phases, numerous studies have demonstrated the potential of AI. For example, Martinez-Amaya et al. (2022) used the random forest algorithm to develop a hurricane prediction framework based on satellite imagery. This framework overcomes the limitations of traditional models in capturing nonlinear features and forecasting rapid intensification. Snezhana (2023) reviewed programs and innovations applied in the mitigation of climate change using AI. Guo et al. (2025) used a DL model, namely, BERT-BiLSTM-CRF, to simulate urban flood spatiotemporal dynamics and influencing factors. They extracted fine-grained flood location and water level data to generate high-resolution flood warnings. Thekdi et al. (2025) examined the compatibility of AI/ML and traditional risk analysis methods in disaster scenarios. Findings shown that the risk community is actively learning and adopting the use of AI/ML-related methods as those methods relate to professional risk-related responsibilities. During the disaster response stage, it is critical to capture and process disaster-related situational information in a timely manner. Huang et al. (2020) used two convolutional neural networks (Inception-V3 and Word Embedding) to extract visual and textual features from social media posts, enabling automatic multimodal classification. This approach provided near real-time documentation of disaster scenarios. Z. Wang et al. (2022) compared the performance of multiple ML methods in detecting rescue requests on social media platforms. These methods included conventional classifiers, ensemble techniques, kernel-based methods, and DL architectures. Li and Wang (2025) addressed the challenges of regional risk uncertainty and complex obstacles in urban flood rescue path planning through a deep reinforcement learning algorithm that integrates mechanisms such as double-priority experience replay and backtracking punishment. It achieved optimal rescue path planning in multi-scale simulation scenarios. H. Zhang et al. (2024) integrated a DL method and a stochastic optimization model to proactively decide optimal quantities of requested resources by considering both currently unfulfilled demands and future demands. Regarding post-disaster recovery stage, Ghaffarian and Emtemani (2021) proposed a ML method combining support vector machine (SVM) and dense conditional random field (DenseCRF). This method aimed to develop monitoring and evaluation technology for urban poor areas, which was unrestricted by geographical location or disaster type. Ro et al. (2024) developed an integrated framework to automatically label and classify massive post-disaster image data. The framework combines pre-trained computer vision models, natural language processing techniques, and disaster ontology. It provides a reusable general paradigm for disaster data management, enabling efficient governance and analytical processes. Sun et al. (2020) provided an overview of applications

of AI in disaster management during its four phases: Mitigation, preparedness, response, and recovery.

Some scholars also combined crowd and AI in disaster management. Fan et al. (2021) developed a digital twin paradigm for disaster response and management by integrating artificial and human intelligence. Wei et al. (2022) proposed a model that can detect false news in social media posts during disasters by combining crowd judgments and ML forecasts. A Bayesian aggregation method was introduced to aggregate the crowd and machine results. Cheng et al. (2023) developed a crowd-AI framework that leveraged the power of AI with citizen science to increase the consistency of post-disaster damage assessment.

Although AI has shown advantages in the complete period of disaster management, its potential drawbacks cannot be ignored. On the one hand, algorithmic bias may lead to an imbalance in resource allocation, such as ignoring the special needs of vulnerable groups such as the elderly and the disabled in the design of evacuation plans; on the other hand, in the face of complex coordination tasks such as cross-departmental collaboration and conflict resolution, AI cannot replace human experience judgment and value trade-off. Several studies have attempted to explore the integration of crowd intelligence with AI, yet they framed the two within a sequential phase-based framework, such as using machine-generated results as input for crowd decision-making and vice versa. The key gap lies in the lack of a comprehensive framework that synchronously integrates crowd and machine intelligence for real-time decision-making in humanitarian response situations. Current approaches do not adequately address the need for a consensus-building model that leverages both types of intelligence to enhance decision-making processes.

2.3 | GDM in Disaster Management

GDM theory is a key paradigm in many disaster management practices, precisely because of the complexity inherent in sudden emergencies—especially large-scale disaster events. Such disasters are characterized by severe hazard intensity, extensive spatial impact, and the need for integrated multi-disciplinary expertise. As a solution for collaborative decision-making, GDM adapts to diverse research perspectives and application contexts, resulting in a rich array of approaches tailored to specific scenarios.

Regarding preference information of experts, scholars focused on breaking through the bottleneck of semantic expressions and consensus reaching. For instance, Zhu et al. (2021) proposed a linguistic preference relationship with confidence level. This method promotes group consensus and enables alternative selection through three key components: Innovative operation rules, a confidence index aggregation method, and a two-step feedback adjustment mechanism. Y. Wang et al. (2022) developed a linguistic preference-based individualized consensus model using consistency-driven optimization to generate personalized interval scales. By incorporating Hausdorff distance metrics and iterative feedback mechanisms, this method resolved significant semantic divergence and low consensus cohesion among experts in large-scale emergency decision-making. Yan et al. (2025) proposed a fuzzy quantum GDM model and demonstrated its

effectiveness through a case study regarding emergency alternative selection for meteorological disasters. H. Wang et al. (2025) developed a large-scale group GDM model based on interval-valued two-tuple linguistic evaluation and used it in location selection for disaster recovery centers.

Some scholars took disaster management as a multi-criteria GDM problem. For example, S. Liu et al. (2022) developed an extended multi-criteria GDM method incorporating psychological factors and bidirectional criteria relationships to address emergency medical supplier selection problems. This approach integrates the Interpretive Structural Model and best-worst method to determine criteria weights. X. Wang et al. (2022) integrated the Swiss cheese model, fault tree analysis and other methods to construct an aviation emergency rescue risk assessment system. He et al. (2024) used multi-source heterogeneous data and GDM to achieve a scientific assessment of urban flood resilience. In addition, the application of integrated methods also opens up new paths for emergency decision-making research. Fallah et al. (2023) proposed a robust optimization framework for mixed swarm decision-making. The framework integrates *k*-means clustering and Borda notation to improve the performance of disaster supply chain networks. Y. Liu et al. (2022) integrated statistical dependency modeling, nonlinear aggregation, and linguistic fuzzy theory to address criteria ambiguity and information uncertainty in online public opinion emergency decision-making. This approach establishes theoretical foundations for priority-based emergency responses under resource constraints. X. Wu and Liao (2024) proposed a multi-criteria GDM model based on alternative chain and trust radius of experts for multi-stage disaster chain scenarios.

Nevertheless, in the current digital era characterized by crowd-sourced intelligence and AI, a significant gap remains. Existing GDM models in disaster management predominantly operate with a pre-defined, static set of alternatives provided by a closed group of experts. There is a lack of a holistic framework that dynamically integrates the real-time, large-scale outputs of crowds and AI to form and evaluate the set of feasible alternatives through stochastic optimization or similar techniques. This study aims to bridge this gap.

3 | Crowd and AI Solutions

Consider a disaster that occurs in a region. To respond to it, a humanitarian organization must assess the number of people in need of rescue in different affected areas and make response decisions. In this section, we first obtain crowd and AI estimations based on crowdsourcing posts. Then, the decisions of the crowd and AI are obtained using the stochastic optimization model considering the uncertain nature of estimations.

3.1 | Crowd Estimations

We first collected disaster relief posts through a disaster crowdsourcing platform. These posts, uploaded by disaster victims, mainly include help-seeking information, addresses, contact details, etc. After obtaining initial data, we need to perform pre-processing, which mainly consists of two steps. First, we delete

duplicates and posts irrelevant to help-seeking information. Second, since the crowdsourcing platform provides upload channels for both victims' help-seeking information and volunteers' relief information, some volunteers mistakenly upload relief information to the help-seeking channel. Therefore, we need to identify and remove these misposted entries.

After obtaining the processed data, we need to count the number of people in need of rescue. To do this, we count the number of vulnerable people in need of rescue by township and finally standardize it by geographical unit. The geocoded rescue posts are tabulated by township and normalized per 10,000 people. In this way, crowd estimations are obtained.

3.2 | AI Estimations

Feature selection should consider the vulnerability of a population to disasters and the population's ability to recover from the impact of disasters. Socio-economic status is an important aspect of increased social vulnerability, which is directly related to the availability of resources and thus affects the impact of coping with disasters (Flanagan et al. 2011; Fatemi et al. 2017). Families include the elderly, children, women, etc., and any of these groups are more likely to require financial support, medical care, or assistance with daily activities during a disaster (Flanagan et al. 2011). Basic public infrastructures, such as schools, fire stations, public transport, and public water systems and pipes, can affect the vulnerability of populations to disasters and emergencies (Cutter et al. 2000; Holand et al. 2011). This study focuses on floods. Thus, we consider water conservancy and embankment engineering reservoirs. Most of the existing studies use statistical indicators to select social vulnerability features, while ignoring the real-time situation of flood disasters.

To improve inclusiveness and compassion in disaster response, we also consider the real-time suffering level of victims in the vulnerability feature system through measuring the urgency level of post textual content. The urgency level of some of the crowdsourcing posts has been marked by rescue organizations in the platform. These marked posts are divided into three levels of urgency with different colors. The highest level of urgency is marked as Level 3 in red; the least urgency is marked as Level 1 in green. For the remaining posts that are not marked, we use the Bidirectional Encoder Representations from Transformers (BERT) model for text multi-classification. BERT is a pre-trained language model based on the transformer architecture. In a text classification task, BERT learns the deep semantic features of text data, maps text to fixed-length vector representations, and then uses these vectors to classify to identify the text's thematic or emotional tendencies.

The features included in these four themes are shown in Table 1. "Positive" indicates that an increase in a feature's value will increase the degree of vulnerability; "Negative" indicates that an increase in a feature's value will reduce the degree of vulnerability.

We used the random forest algorithm to fit 16 features in the social vulnerability system to predict the number of people in need of rescue. We adopt the percentage ranking method (Flanagan et al. 2011) to nondimensionalize these features. The percentile rank is

TABLE 1 | Social vulnerability features.

Theme	Variable	Contribution
Real-time degree of human suffering	The urgency level	Positive
Socioeconomic status	The proportion of minimum livelihood protection	Positive
	Per capita disposable income	Negative
	Number of 5G base stations	Negative
	Number of industrial enterprises	Negative
	Number of supermarkets with a sales area of over 50 m ²	Negative
Household composition	The proportion of persons aged 65 and older	Positive
	The proportion of persons aged 14 and younger	Positive
	The proportion of female population	Positive
Hydraulic and levee engineering	Number of reservoir projects	Negative
	Number of hydroelectric power stations	Negative
	Number of pumping stations	Negative
	Length of levees	Negative
	Population protected by levees	Negative
	Length of river sections with flood control responsibilities	Positive
	Drainage area	Negative

defined as the proportion of scores in a distribution that are less than or equal to a specific score. The formula for calculating the percentile rank is as follows:

$$\text{Percentile rank} = \frac{(\text{Rank} - 1)}{(N - 1)} \quad (1)$$

“Rank” is the ranking of townships for a certain feature, and “N” is the total number of townships. A higher value of Equation (1) corresponds to a higher vulnerability level. For variables with positive contributions, the percentile ranks of townships are calculated by ranking from low to high; for variables with negative contributions, the percentile ranks of townships are calculated by ranking from high to low.

k-fold cross-validation is adopted to partition the data set into *k* equally sized, mutually exclusive subsets. In each training iteration, we use *k* − 1 of these subsets as the training set, and the remaining subset as the test set. This process is repeated *k* times, and the average of the performance metrics returned from the *k* training cycles is the final performance of the model. During training, we selected seven important features based on feature importance. The number of decision trees was determined by *k*-fold cross-validation results using the MSE performance metric.

3.3 | Crowd and AI Decisions

3.3.1 | Scenario Generation and Reduction

Based on the above research, we obtained crowd and AI estimations about the number of vulnerable people in need of rescue. Each source carries significant uncertainty regarding the number of people in need of rescue. Recall the previously mentioned

uncertainty associated with crowd estimates. Uncertainty in machine estimation arises from humanitarian ethical pressures and the nature of the AI used. The humanitarian literature commonly defines probability distributions to represent the uncertainties caused by parameters (i.e., in this study, the number of people in need of rescue) and adopts stochastic optimization to support response decisions (Salmerón and Apte 2010). To deal with this issue, we present a framework for building a stochastic programming model (SP) and extracting crowd and AI decisions.

Stochastic optimization enables a problem to be solved through the formulation of uncertain parameters based on an explicitly defined probability measure. Given any probability measure, a set of scenarios with associated probabilities p_s for $s \in S$ is generated. A scenario represents a realization of the uncertain parameters. First, we provide an approach to generate scenarios.

Corresponding to the interval nature of uncertain estimation for the number of vulnerable people in need of rescue, we adopt triangular probability distributions, consisting of the upper bound, lower bound, and most probable values. The three values are denoted as *UB* (upper bound), *LB* (lower bound), and *BC* (base case). In particular, we use the estimates from the two data sources as *BC*. *UB* and *LB* are generated based on an uncertainty parameter δ , where $UB = BC * (1 + \delta)$ and $LB = BC * (1 - \delta)$. The parameter $\delta \in [0, 1]$ indicates the confidence associated with the estimation. A larger δ corresponds to lower confidence in the estimation, implying greater uncertainty.

Uncertainty levels correlate with demand estimate discrepancies between crowdsourced and machine evaluations. Greater discrepancies correspond to lower confidence in both estimates and higher demand uncertainty. We therefore formulate the uncertainty parameter probability distribution based on this

TABLE 2 | Probability distribution for δ .

< 1						$\text{con}_i \geq 1$			
δ	0.1	0.2	0.3	0.4	0.5	δ	0.2	0.3	0.5
Probability	0.1	0.2	0.4	0.2	0.1	Probability	0.3	0.4	0.3

discrepancy. Let q_i^C and q_i^A denote the crowd and AI evaluations at affected site i , respectively. The discrepancy metric is defined as follows.

$$\text{con}_i = \frac{\max \{q_i^C, q_i^A\} - \min \{q_i^C, q_i^A\}}{\min \{q_i^C, q_i^A\}} \quad (2)$$

The parameter δ probability distribution is shown in Table 2.

In general, an accurate representation of possible realizations requires numerous scenarios. Accordingly, the computational burden will be heavy in the face of many scenarios. We need to adopt a scenario reduction process to manage the computational burden. We therefore apply a clustering process to cluster scenarios into different clusters. k -means clustering algorithm is employed for its simplicity and computational efficiency, particularly when scaling to large datasets. Determining the optimal cluster count k is critical. We utilize the elbow method by analyzing the rate of change in sum of squared errors (SSE) versus k values. The inflection point where SSE reduction substantially slows indicates optimal k .

3.3.2 | The Optimization Model

After we obtain the scenarios, our next task is to construct the stochastic optimization model. Table 3 provides the model notation.

The model is formulated as follows:

Objective function

$$\max \sum_{s \in S} P_s \cdot \left(\sum_{t \in T} \sum_{a \in A} R_{t,a,s} \right) \cdot \exp(-\mu_d \cdot D_s(x) - \mu_r \cdot P_s(x)) \quad (3)$$

First-stage constraints

$$\sum_{o \in O} w_{t,o} = z_t, \forall t \in T \quad (4)$$

$$1 \leq \sum_{t \in T} w_{t,o} \leq K, \forall o \in O \quad (5)$$

$$\sum_{t \in T} \sum_{o \in O} w_{t,o} = \sum_{t \in T} z_t \leq K \quad (6)$$

Second-stage constraints

Path continuity constraints

$$\sum_{a' \in A} x_{t,a',a,s} = \sum_{a' \in A} x_{t,a,a',s} = y_{t,a,s} \quad (7)$$

$$\sum_{a \in A} x_{t,o,a,s} = \sum_{a \in A} x_{t,a,o,s} = w_{t,o} \quad (8)$$

$$\sum_{o \in O} \sum_{a \in A \setminus \{o\}} x_{t,o,a,s} = \sum_{o \in O} \sum_{a \in A \setminus \{o\}} x_{t,a,o,s} = z_t \quad (9)$$

Time flow constraints

$$A_{t,a',s} \geq F_{t,a,s} + t_{a,a'} - M(1 - x_{t,a,a',s}) \quad (10)$$

$$F_{t,a,s} = S_{t,a,s} + \alpha \cdot R_{t,a,s} \quad (11)$$

$$S_{t,a,s} \geq A_{t,a,s} \quad (12)$$

if

$$w_{t,o} = 1, A_{t,o,s} = 0 \quad (13)$$

Team capacity constraints

$$\sum_{a \in A} R_{t,a,s} \leq h_t \cdot z_t \quad (14)$$

$$\sum_{a \in A} R_{t,a,s} \geq z_t \quad \forall t \in T, s \in S \quad (15)$$

Variable domain constraints

$$z_t, w_{t,o} \in \{0, 1\}, \quad \forall t \in T, o \in O \quad (16)$$

$$x_{t,a,a',s}, y_{t,a,s} \in \{0, 1\}, \quad \forall t \in T, a, a' \in A, s \in S \quad (17)$$

$$R_{t,a,s}, A_{t,a,s}, F_{t,a,s}, S_{t,a,s} \geq 0, \quad \forall t \in T, a \in A, s \in S \quad (18)$$

The objective of Equation (3) is to maximize the expected number of people rescued, while incorporating penalty terms related to time window violations and unmet demand via an exponential function to balance the timeliness and coverage of rescue operations. Equations (4)–(18) serve as the model constraints, which are divided into first-stage (scenario-independent) and second-stage (scenario-dependent) constraints. Constraint (4) indicates that

TABLE 3 | Model notation.

Notation	Meaning
Sets	
T	Set of rescue teams, $t \in T$
A	Set of areas to be rescued, $a \in A$, each area corresponds to a Cartesian coordinate (x_a, y_a)
O	Set of rescue starting points, $o \in O$, $O \subseteq A$ (coincides with the rescue areas), with coordinates (x_o, y_o)
S	Set of rescue scenario, $s \in S$
Scenario-independent parameters	
p_s	Probability that scenario s occurs, satisfying $\sum_{s \in S} p_s = 1$
$vul_{a,s}$	Number of people to be rescued in area a under scenario s
Spatial and temporal parameters	
(x_a, y_a)	Cartesian coordinates of the area a to be rescued
$d_{a,a'}$	Euclidean distance from area a to area a' where $d_{a,a'} = \sqrt{(x_a - x_{a'})^2 + (y_a - y_{a'})^2}$
v	Travel speed of rescue teams (it is assumed that all 42 teams have the same speed, $v = 30$ km/h)
$t_{a,a'}$	Travel time of the team from area a to area a' , where $t_{a,a'} = d_{a,a'}/v$ (unit: hours)
$[e_a, l_a]$	Rescue time window for area a , where e_a and l_a represent the earliest arrival time and the latest arrival time, respectively
Resource and demand parameters	
h_t	Maximum rescue capacity of team t
K	Number of rescue teams
γ	Lower bound of demand satisfaction rate $\gamma = 0.7$, ensuring that more than 70% of people to be rescued are covered)
u_d	Penalty coefficient for time window constraints
u_r	Penalty coefficient for demand constraints
First-stage variables	
z_t	Whether to dispatch team t ($z_t = 1$ indicates dispatch)
$w_{t,o}$	Whether team t departs from starting point o
Second-stage variables	
$x_{t,a,a',s}$	Under scenario s , whether team t travels from area a to area a'
$y_{t,a,s}$	Under scenario s , whether team t visits area a
$R_{t,a,s}$	Under scenario s , number of people rescued by team t in area a
$A_{t,a,s}$	Under scenario s arrival time of team t at area a
$S_{t,a,s}$	Under scenario s , start time of rescue by team t in area a
$F_{t,a,s}$	Under scenario s , completion time of rescue by team t in area a

each dispatched rescue team must correspond to exactly one starting point. Constraint (5) states that the number of teams departing from each starting point should stay within a reasonable range. Constraint (6) ensures that the total number of starting point selections aligns with the number of dispatched teams, and this number does not exceed the total number of available teams. Path continuity constraints (7)–(9) maintain the logical consistency of rescue routes: Constraint (7) ensures that the number of incoming and outgoing routes for a team at any area matches its visit status to that area; Constraint (8) links the route of a team to its selected starting point; Constraint (9) guarantees that only dispatched teams have valid route arrangements. Time flow constraints (10)–(13) regulate the temporal logic of rescue operations. Constraint

(10) defines that a team's arrival time at an area is no earlier than the completion time of its previous rescue plus the travel time between areas; Constraint (11) clarifies that the rescue completion time at an area equals the start time plus the duration determined by the number of people rescued; Constraint (12) ensures the rescue start time is not earlier than the arrival time at the area; Constraint (13) sets the arrival time at the selected starting point to the initial moment (zero). Team capacity constraints (14)–(15) limit the rescue scale of each team: Constraint (14) states that the total number of people rescued by a team cannot exceed its maximum capacity; Constraint (15) requires that any dispatched team must undertake at least one rescue task. Finally, Constraints (16)–(18) define the variable domains

Using the obtained scenarios to solve (3–18), we can obtain a series of solutions $a(s)$ for the crowd and AI scenarios.

4 | Consensus Building Based on Maximum Consensus Sequence

In real-world disaster emergency decision-making, although optimization models are often used as references, organizational experts still make the final decisions based on their governance experience. In light of this reality, we construct a GDM model with the crowd and AI solutions. Crowd and AI solutions obtained by the model (3–18) are used as feasible alternatives. Experts provide preference information over these alternatives. Finally, a consensus-building method based on mining maximum consensus sequences is introduced.

$E = \{e_1, e_2, \dots, e_n\}$ represents a collection of n experts from related disaster fields and $A = \{a_1, a_2, \dots, a_m\}$ represents a set of m alternatives, which is generated from Model (3–18). We use reciprocal preference relations (RPRs) (Świtalski 1999) to express experts' preferences. An RPR $P(e_k) = (p_{ij}(e_k))_{m \times m}$ meets the following conditions:

$$p_{ij}(e_k) \in [0, 1], p_{ij}(e_k) + p_{ji}(e_k) = 1, p_{ij}(e_k) = 0.5, \forall i, j \in 0, 1, \dots, n$$

$p_{ij}(e_k)$ represents the degree of preference for alternative a_i over a_j . When $p_{ij}(e_k) > 0.5$, it means that the expert prefers alternative a_i over a_j . When $p_{ij}(e_k) < 0.5$, it means that the expert prefers alternative a_j over a_i . When $p_{ij}(e_k) = 0.5$, it means that the expert has equal preference between alternative a_i and a_j .

To comprehensively capture the complex nature of disasters, an emergency decision-making problem usually involves experts from different departments. These experts usually have varied perspectives. Thus, it may be difficult to achieve an expected consensus. This section aims to mining potential consensus sequences from the preferences of experts and connecting these sequences and perspectives. Thus, the sequence preference $spu_{ij}(e_k)$ of e_k for an alternative pair $a_i > a_j$ is defined as follows.

$$spu_{ij}(e_k) = p_{ij}(e_k) \quad (19)$$

Based on Equation (19), the ranking preference $rp(e_k)$ for $a_1 > a_2 > \dots > a_m$ is defined as follows.

$$rp(e_k)(a_1 > a_2 > \dots > a_m) = \frac{spu_{a_1 a_2}(e_k) + spu_{a_2 a_3}(e_k) + \dots + spu_{a_{m-1} a_m}(e_k)}{m-1} \quad (20)$$

In a group, the degree of consensus for a sequence $a_1 > a_2 > \dots > a_i$ is defined based on Equation (20) as follows.

Where $w(e_k)$ is the weight of e_k .

$$cs(a_1 > a_2 > \dots > a_i) = \sum_{k=1}^n w(e_k) \cdot rp(e_k)(a_1 > a_2 > \dots > a_i) \quad (21)$$

Let B be the consensus threshold, if the degree of consensus of a sequence satisfies $cs(a_1 > a_2 > \dots > a_i) \geq ct$, then this sequence $a_1 > a_2 > \dots > a_i$ is called a consensual sequence.

The optimal support for an alternative a_i is defined based on Equation (21) as $opa(e_k)(a_i)$:

$$opa(e_k)(a_i) = \frac{1}{m-1} \sum_{j=1, j \neq i}^m spu_{a_i a_j}(e_k)(a_i > a_j) \quad (22)$$

The optimal consensus support for an alternative a_i is defined based on Equation (22) as $oca(a_i)$:

$$oca(a_i) = \sum_{k=1}^n w(e_k) \cdot opa(e_k)(a_i) \quad (23)$$

If the maximum consensual sequence does not include all alternatives, it is necessary to improve the consensus sequence. After declaring the above definitions (19)–(23), we propose the maximum consensus sequence mining algorithm as follows.

A sequence with length q is regarded as a q -sequence. Let L_q denote the set of all q -consensus sequences, and C_q denote the set of candidate q -consensus sequences.

Algorithm: The consensus-building process

Input: Experts' RPRs, consensus threshold ct , and the maximum number of permitted iterations max .

Output: Complete consensus sequence; total number of iterations.

Step 1: L_1 is the set of alternatives.

Step 2: Iterate the following steps to generate the consensus sequence set L_q .

for ($q = 2, L_{q-1} \neq \emptyset, q++$) do

$$C_q = geneL_q(L_{q-1})$$

$$L_q \{ C \in C_q | cs(c) \geq ct \}$$

We input the initial sequence set L_1 and generate C_2 through operating $geneL_2(L_1)$. Then, we calculate the degree of consensus of sequences in C_2 . Sequences with a degree of consensus greater than ct are put into subset C_q , and this process is repeated iteratively to finally generate the sequence set C_q . The generation of C_q depends on L_{q-1} since a q -sequence can become a consensus sequence only if its $(q-1)$ sub-sequences are consensus sequences (Chen and Cheng 2009).

Procedure $geneL_q(L_{q-1})$: for each sequence $l_1, l_2 \in L_{q-1}$ if ($q = 2$), then

Generate candidate C_2 else if the last $q-2$ alternatives in l_1 are equal to the first $q-2$ alternatives in l_2 , then join l_1 with l_2 to generate a candidate a sequence and add it to C_q

Return C_q

For example, if sequence $l_1 = \{a_1 > a_3 > a_2\}$ and sequence $l_2 = \{a_2 > a_5 > a_4\}$, the last scheme of l_1 is the same as the first scheme of l_2 , then merge l_1 and l_2 to generate a new sequence $\{a_1 > a_3 > a_2 > a_5 > a_4\}$.

Step 3: Remove those sequences that are included in other sequences from the L_q sequence set to find the largest consensus sequence among them.

Step 4: If $q = m$, the maximum consensus sequence obtained by **Step 3** is the final consensus sequence. If $q < m$, we need to do the following steps.

Step 4.1: Calculate the optimal consensus support for each alternative and sort these alternatives in descending order according to $oca(a_i)$ to get the potential consensus ranking.

Step 4.2: Compare the potential consensus ranking with the maximum consensus sequence to find the alternative pair that breaks the complete sequence.

Step 4.3: Identify experts who need to modify preferences: $EXP = \{(a_i, a_j) | p_{ij}(e_k) < ct\}$. These experts accept the revision and increase the preference regarding $p_{ij}(e_k)$ (we assume that an expert will add 0.15 each time to the original score).

Step 4.4: Recalculate degree of the consensus of the modified sequence. If the degree of consensus is no less than ct or $it > max$, the iteration is ended; Otherwise, $it = it + 1$, return to **Step 4.1** to continue iterating.

5 | The Procedure of the Proposed Model

The complete procedure of the proposed model is summarized below and presented in Figure 2.

Step 1: Obtain crowd estimation of rescue needs. Crowdsourcing posts submitted by victims on a crowdsourcing platform are collected. The number of vulnerable people in need of rescue in each text is counted to obtain an estimate of the number of vulnerable people in the crowd.

Step 2: Construct the vulnerability feature system. The urgency level is also determined using the BERT to obtain one feature theme, namely, the real-time degree of human suffering.

Step 3: Obtain AI's estimate of rescue needs. Use an ML algorithm to estimate the number of vulnerable people in need of rescue.

Step 4: Scenario generation and reduction. Generate scenarios using a triangular probability distribution and reduce scenarios using a clustering algorithm.

Step 5: Build a stochastic optimization model to obtain the crowd and AI solutions.

Step 6: Collect expert preferences and use the consensus-building process through mining the maximum consensus sequences to reach a consensus for GDM.

6 | Case Study

From July 20 to 23, 2021, Henan Province was hit by extremely heavy rains, causing serious casualties and property losses (Xin-

hua News Agency 2022). During the disaster, the emergence of Tencent's shared documents has provided a bridge for information exchange among victims, humanitarian organizations, and volunteers. Affected individuals can report distress locations and contact details through designated interfaces, while volunteers contributed resource availability data such as relief supplies and shelters. Concurrently, rescue teams could register their own rescue capabilities and communication channels within the integrated platform. To maximize the effectiveness of rescue supplies, we have collected and sorted the demand and supply information in this document, and constructed a rescue team dispatch model. The model aims to deploy rescue forces to the most urgent populations. To achieve this goal, we provide decision-making solutions from the crowd and AI, and organizational experts provide perspectives regarding the solutions. This section uses Xinxiang of Henan Province as an example to illustrate and verify the method proposed in this paper.

Xinxiang City has four districts of Weibin District, Hongqi District, Muye District, and Fengquan District, three county-level cities of Huixian City, Weihui City, and Changyuan City, and five counties of Xinxiang County, Huojia County, Yuanyang County, Yanjin County, and Fengqiu County, with a total of 152 townships. We solve a rescue team dispatch problem. Rescue teams and their rescue capacity are presented in the Appendix (Table A1–A8) (Figure A1–A5).

6.1 | Model Implementation

Step 1: Obtain crowd estimation of rescue needs. We collected a total of 1669 posts from the crowdsourcing platform, deleted 145 invalid posts, and finally obtained 1524 valid posts. Among them, the urgency levels of 537 posts were labeled by rescue workers, while the urgency levels of the remaining 987 posts were unknown. Then, we calculate the number of vulnerable individuals needing rescue per township from 1524 posts (see Figure 3). During the disaster, this crowdsourcing platform was widely used and covered multiple affected cities in Henan Province. It documented thousands of pieces of real-time assistance requests and rescue-related information, demonstrating its broad coverage and timeliness. The information was structured uniformly with key fields such as location and demand type, which systematically captures typical needs and resource distribution in disaster rescue operations. Furthermore, this data source has been cited in multiple media reports and adopted in subsequent studies, indicating its reliability and representativeness as a high-quality crowdsourced disaster data.

Step 2: Construct the vulnerability feature system. Sixteen features are selected to establish a social vulnerability index system. The data for the 16 characteristic variables across the streets and townships of Xinxiang City were obtained from multiple authoritative public sources. The proportion of persons aged 65 and older, aged 14 and younger, female population, and per capital disposable income were obtained from the Seventh National Population Census. The proportion of minimum livelihood protection was collected from publicly available documents published by the district governments of Xinxiang. Data on the number

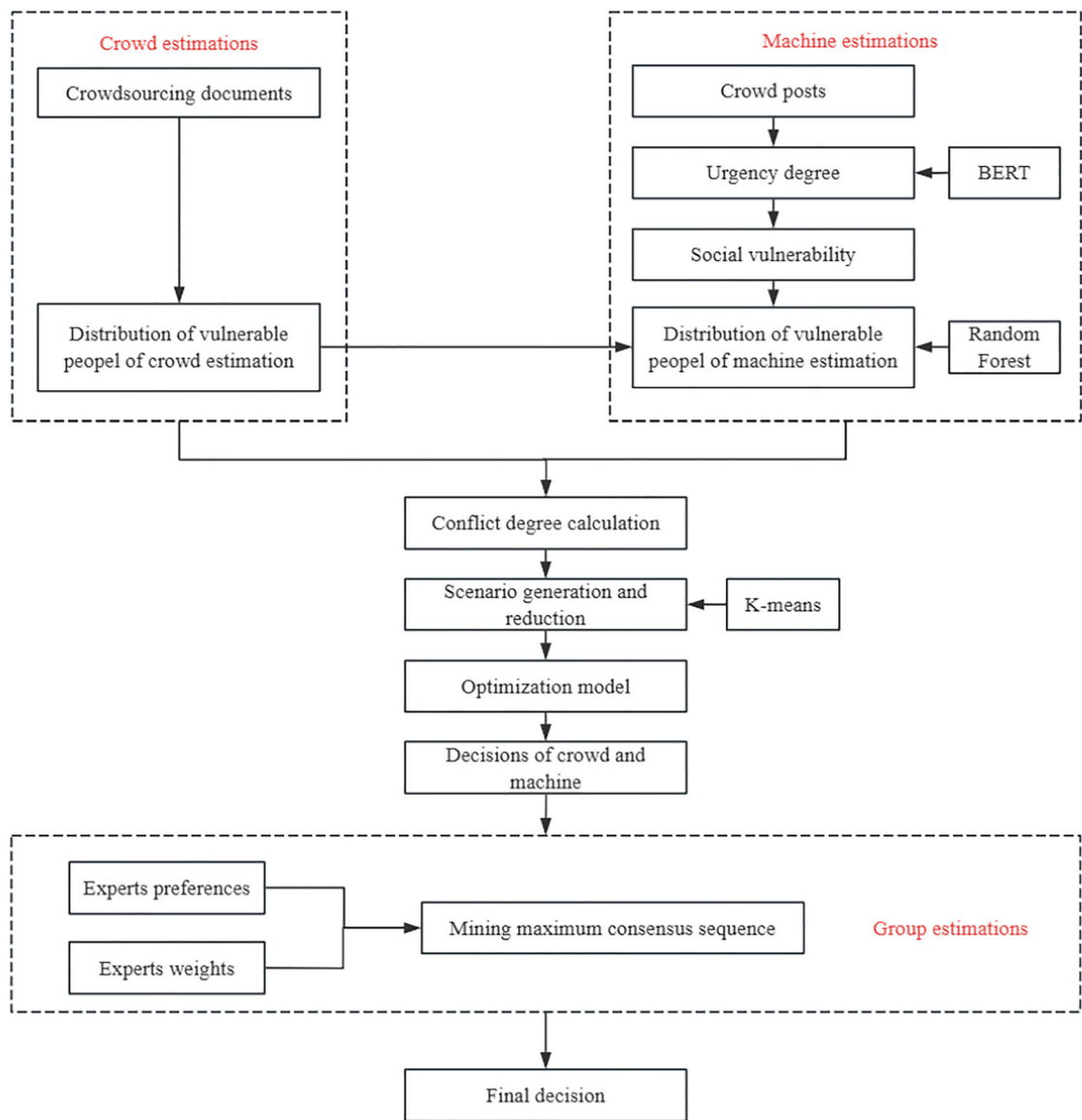


FIGURE 2 | The procedure diagram.

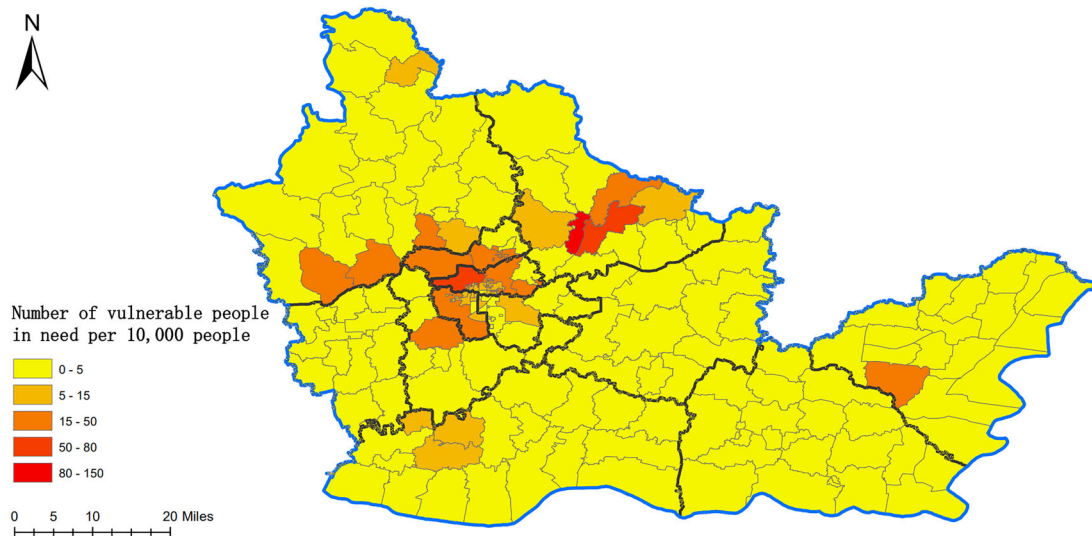


FIGURE 3 | Crowd distribution.

TABLE 4 | Performance metrics of various types for urgency classification task (BERT).

Class	Precision	Recall	F1
Level 1	74.12%	85.14%	79.25%
Level 2	82.11%	70.27%	75.73%
Level 3	80.90%	85.71%	83.24%

of 5G base stations, industrial enterprises, supermarkets with a sales area of over 50 m², reservoirs, pumping stations, and drainage area were extracted from the China County Statistical Yearbook (2022). Furthermore, data on length of levees and the population protected by levees were integrated from the 2025 flood control project records of the Zhengzhou Water Authority and the subregional statistics of the Seventh National Population Census. We used the BERT model to measure the real-time urgency level. Table 4 presents the precision, recall, and F1-score for the classification task of the urgency level.

In addition, we compared BERT with two baseline models (logistic regression and SVM) (see Table 5).

The experimental results show that our BERT model outperforms LR model by 3.84% and SVM model 3.88% in terms of F1-score. While an improvement of approximately 3.8% may appear numerically modest, it holds substantial practical significance in emergency disaster response. In practical terms, this margin translates to accurately identifying dozens of additional critical rescue posts that baseline models miss due to their reliance on explicit keywords. Since a single post often involves multiple vulnerable individuals, minimizing false negatives is paramount for life-saving efforts. Furthermore, although BERT requires more computational resources for training than LR or SVM, its inference time is virtually negligible on modern hardware. In high-stakes emergency scenarios, the life-saving potential of BERT's superior semantic understanding heavily outweighs the marginal increase in computational overhead. These comparative results demonstrate that, given the complexity and data characteristics of the classification task, the BERT model has a better performance.

Step 3. Obtain AI's estimate of rescue needs. Seven important features are selected and fitted by random forest to obtain the estimated number of vulnerable people in need of rescue (see Figure 4). In the feature selection stage, we conducted ablation experiments on the "urgency level" feature (see Table 6). The results indicate that the demand prediction error deteriorates significantly after removing this feature: MSE increases by 146.71%, MAE rises by 39.07%, and the coefficient of determination R^2 decreases by 31.33%, respectively. This result verifies the critical role of the urgency level in ensuring the accuracy of demand prediction, and further corroborates the necessity and rationality of our previous operation of classifying help-seeking texts by urgency level using the BERT model.

In addition, we compared the random forest model with two transparent linear baseline models (linear regression and Ridge regression) (see Table 7). The experimental results show that the used random forest model outperforms these two linear baseline models in prediction performance, which proves the necessity of considering nonlinear relationships and interaction effects between features when addressing such complex prediction problems.

Step 4: Scenario generation and reduction. Calculate the crowd-AI conflict level regarding their estimation results. Then, 200 scenarios are randomly generated. For each source, each scenario S is assigned with the same probability, that is, $p^s = 0.005$. All scenarios are clustered by the k -means clustering algorithm. Based on the clustering result (see Table 8), the numbers of clusters for the crowd and AI are 3 and 2 (see Figures 5 and 6), respectively. Thus, GDM will have five feasible alternatives. The details of five alternatives are presented in the Appendix.

Step 5: Build a stochastic optimization model to obtain the crowd and AI solutions. The reduced scenario obtained by Step 4 is substituted into the stochastic optimization model, and five decision-making alternatives of the crowd and the AI are obtained.

Step 6: Collect expert preferences. Five experts are invited to evaluate five feasible alternatives. RPRs of five experts from relevant government agencies and organizations are shown in Table 9.

The weights of five experts are set as $w(e_1) = 0.267$, $w(e_2) = 0.267$, $w(e_3) = 0.133$, $w(e_4) = 0.133$, $w(e_5) = 0.200$.

We assume that the consensus threshold $ct = 0.65$. We first generate $L_1 = \{x_1, x_2, x_3, x_4, x_5\}$ and generate C_2 from $geneL_2(L_1)$. Find the consensus threshold greater than 0.65 in all sequences of C_2 , and divide them into L_2 . The resulting consensus for all 2-sequences is as follows.

$$css(x_1 > x_4) = 0.753, css(x_2 > x_1) = 0.673, css(x_2 > x_4) = 0.720, css(x_3 > x_1) = 0.714,$$

$$css(x_3 > x_2) = 0.687, css(x_3 > x_5) = 0.653.$$

Therefore, $C_2 = \{x_1 > x_4, x_2 > x_1, x_2 > x_4, x_3 > x_1, x_3 > x_2, x_3 > x_5\}$. Repeat the steps above to generate C_3 based on C_2 . From this, we get $C_3 = \{x_2 > x_1 > x_4, x_3 > x_1 > x_4, x_3 > x_2 > x_1, x_3 > x_2 > x_4\}$. In the same way, we can get $C_4 = \{x_3 > x_2 > x_1 > x_4\}$, and the maximum consensus sequence is $x_3 > x_2 > x_1 > x_4$. At generation C_5 , the maximum consensus sequence mining algorithm stops when it executes to Step 2. Next, Step 4 of the maximum consensus sequence mining algorithm is executed, and the optimal scheme consensus degree for each scheme is calculated.

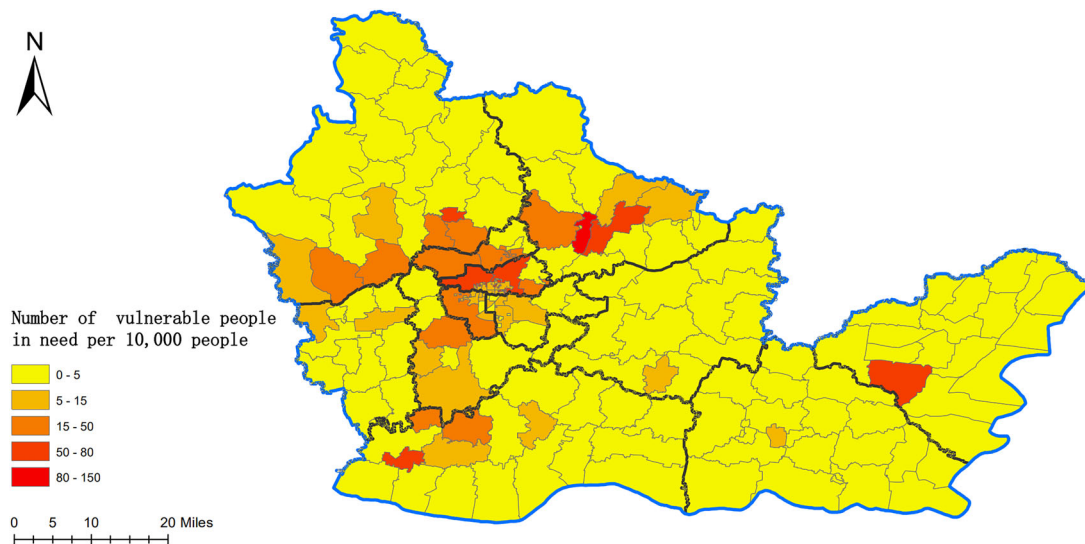


FIGURE 4 | AI distribution.

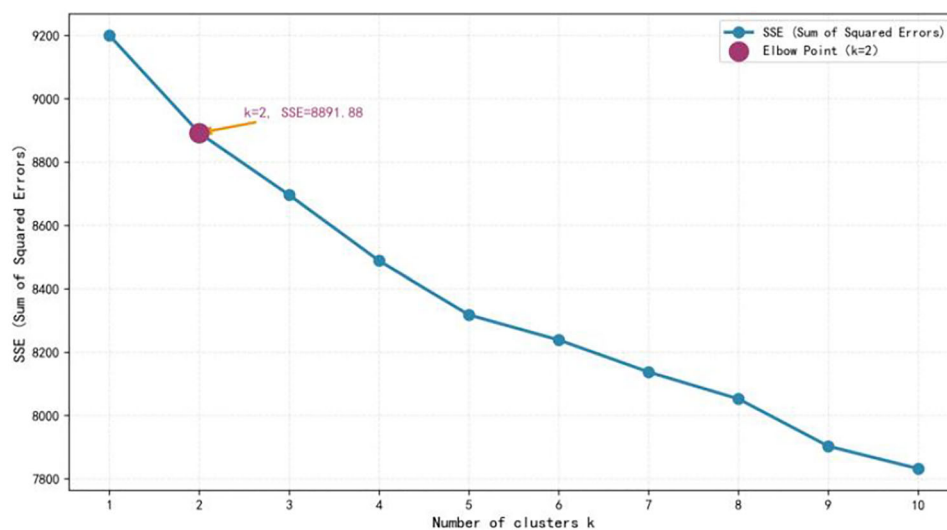


FIGURE 5 | The elbow plot of crowd.

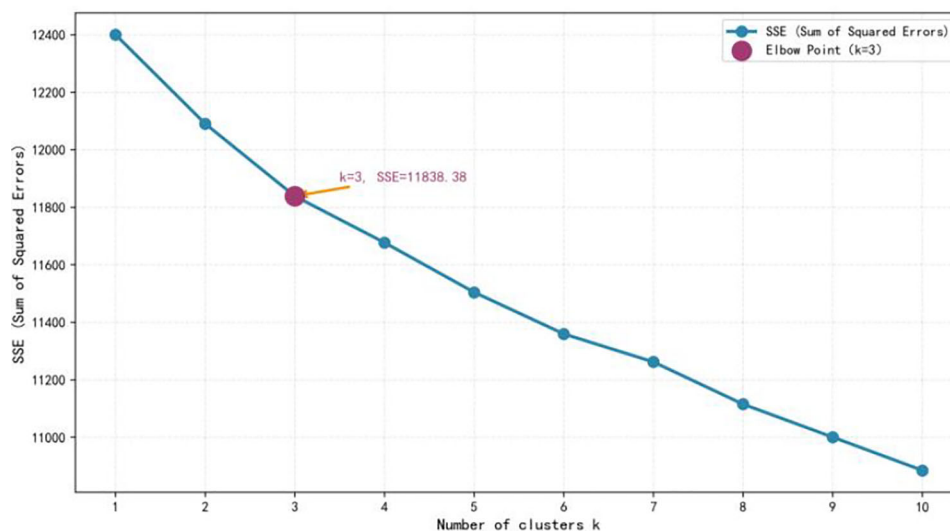


FIGURE 6 | The elbow plot of AI.

TABLE 5 | Comparative experiment of classification task (BERT).

Model	Precision	Recall	F1
Logistic regression	75.46%	75.46%	75.20%
SVM	74.72%	74.72%	75.16%
BERT	79.53%	79.18%	79.04%

TABLE 6 | Ablation experiment on urgency level.

Feature set	Test set MSE	Test set MAE	Test set R^2
7 features (including urgency)	39.7465	2.9739	0.8240
6 features (excluding urgency)	98.0567	4.1359	0.5659

TABLE 7 | Comparative experiment of demand prediction task.

Model	Dataset	MSE	MAE	R^2
Linear regression	Train dataset	38.9797	3.2716	0.4497
	Test dataset	712.3621	9.1678	0.0825
Ridge regression	Train dataset	39.6542	2.9718	0.4402
	Test dataset	703.9033	8.8007	0.0934
Random forest	Train dataset	24.7447	1.2546	0.8860
	Test dataset	39.7465	2.9739	0.8240

TABLE 8 | Clustering result.

	Clusters	Weights
Crowd	Cluster 1	0.470
	Cluster 2	0.530
AI	Cluster 1	0.415
	Cluster 2	0.320
	Cluster 3	0.265

$oca(x_1) = 0.430$, $oca(x_2) = 0.542$, $oca(x_3) = 0.588$,
 $oca(x_4) = 0.340$, $oca(x_5) = 0.575$.

Thus, $oca(x_3) > oca(x_5) > oca(x_2) > oca(x_1) > oca(x_4)$ are obtained. The potential consensus ranking is $x_3 > x_5 > x_2 > x_1 > x_4$. Comparing the sequence $x_3 > x_5 > x_2 > x_1 > x_4$ with $x_3 > x_2 > x_1 > x_4$, it is found that the alternative that breaks the complete chain is x_5 . Since x_5 is in the middle place of the potential consensus sequence, the alternative pairs that need to be modified are (x_3, x_5) and (x_5, x_2) .

$$p_{35}(e_4) = 0.5 < 0.65, p_{35}(e_5) = 0.6 < 0.65$$

$$p_{52}(e_1) = 0.6 < 0.65, p_{52}(e_2) = 0.5 < 0.65, p_{52}(e_5) = 0.3 < 0.65$$

Therefore, it is necessary to improve the evaluation of experts e_4 and e_5 on (x_3, x_5) and the evaluation of experts e_1 , e_2 , and e_5 on (x_5, x_2) . It is assumed that all experts accept the recommendations and make changes accordingly. Recalculate the sequence $css(x_3 > x_5 > x_2 > x_1 > x_4) = 0.695 > ct$. Therefore, the final alternative ranking is ordered as $x_3 > x_5 > x_2 > x_1 > x_4$.

6.2 | Experimental Analysis

6.2.1 | Inclusiveness Analysis

In addition to conventional performance indicators for rescue plans, such as rescue rate and time window violation rate, we introduce the resource matching degree for low-connectivity areas as a key evaluation dimension, guided by the core principle of prioritizing vulnerable groups.

Low-connectivity areas are identified using the normalized count of 5G base stations across all 152 regions in need of rescue. The 25th percentile of this normalized indicator is 0.2366. Areas with a normalized 5G base station count less than or equal to 0.2366 are classified as low-connectivity. These areas typically have fewer 5G base stations originally installed, reflecting weaker communication infrastructure. This classification is consistent with the concept of low-connectivity areas, where disaster-related information is more likely to go unreported due to limited network coverage.

TABLE 9 | Expert preferences.

Expert	Preference matrix
e_1	$P(e_4) = \begin{bmatrix} 0.5 & 0.7 & 0.3 & 0.5 & 0.3 \\ 0.3 & 0.5 & 0.4 & 0.1 & 0.4 \\ 0.7 & 0.6 & 0.5 & 0.8 & 0.3 \\ 0.5 & 0.9 & 0.2 & 0.5 & 0.5 \\ 0.7 & 0.6 & 0.7 & 0.5 & 0.5 \end{bmatrix}$
e_2	$P(e_4) = \begin{bmatrix} 0.5 & 0.8 & 0.2 & 0.7 & 0.4 \\ 0.2 & 0.5 & 0.3 & 0.3 & 0.5 \\ 0.8 & 0.7 & 0.5 & 0.9 & 0.3 \\ 0.3 & 0.7 & 0.1 & 0.5 & 0.4 \\ 0.6 & 0.5 & 0.7 & 0.6 & 0.5 \end{bmatrix}$
e_3	$P(e_4) = \begin{bmatrix} 0.5 & 0.6 & 0.4 & 0.6 & 0.3 \\ 0.4 & 0.5 & 0.5 & 0.4 & 0.3 \\ 0.6 & 0.5 & 0.5 & 0.2 & 0.3 \\ 0.4 & 0.6 & 0.8 & 0.5 & 0.3 \\ 0.7 & 0.7 & 0.7 & 0.7 & 0.5 \end{bmatrix}$
e_4	$P(e_4) = \begin{bmatrix} 0.5 & 0.6 & 0.6 & 0.7 & 0.5 \\ 0.4 & 0.5 & 0.3 & 0.3 & 0.7 \\ 0.4 & 0.7 & 0.5 & 0.5 & 0.5 \\ 0.3 & 0.7 & 0.5 & 0.5 & 0.6 \\ 0.5 & 0.3 & 0.5 & 0.4 & 0.5 \end{bmatrix}$
e_5	$P(e_5) = \begin{bmatrix} 0.5 & 0.7 & 0.1 & 0.7 & 0.3 \\ 0.3 & 0.5 & 0.1 & 0.4 & 0.7 \\ 0.9 & 0.9 & 0.5 & 0.5 & 0.4 \\ 0.3 & 0.6 & 0.5 & 0.5 & 0.6 \\ 0.7 & 0.3 & 0.6 & 0.4 & 0.5 \end{bmatrix}$

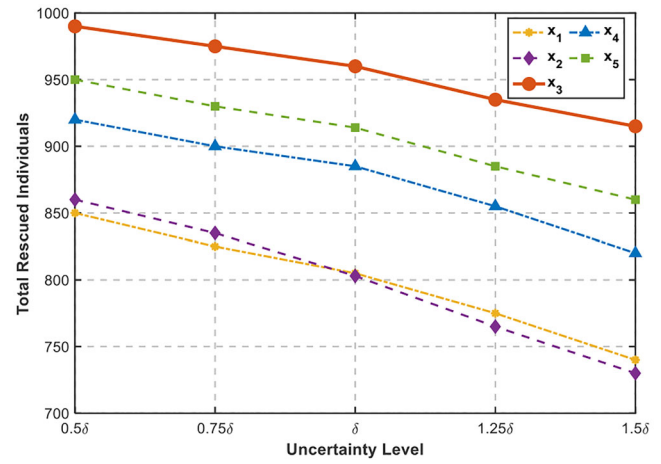
The resource matching degree for low-connectivity areas is formulated as:

$$R_{\text{low}} = \frac{N_{\text{rescued,low}}}{N_{\text{rescued,total}}} / \frac{D_{\text{low}}}{D_{\text{total}}}$$

where $N_{\text{rescued,low}}$ denotes the total number of people rescued in low-connectivity areas, $N_{\text{rescued,total}}$ denotes the total number of people rescued in all areas, D_{low} denotes the total number of people rescued in need of rescue in low-connectivity areas, and D_{total} denotes the total number of people in need of rescue in all areas. If $R_{\text{low}} \leq 1$, values closer to 1 indicate a better match between allocated rescue resources and actual demand in low-connectivity areas. If $R_{\text{low}} > 1$, it reflects a deliberate allocation of additional resources toward low-connectivity areas, consistent with the principle of ensuring fairness by prioritizing vulnerable groups in rescue operations.

After comparing the metrics of each rescue plan (see Table 10), we find that Scenario x_3 performs well across multiple key indicators, especially in terms of rescue efficiency and fairness in resource allocation.

Specifically, despite facing the largest number of people in need of rescue (1117 individuals), x_3 achieved a rescue rate of 85.94%, demonstrating strong emergency response and dispatching capability. Compared to other machine-based scenarios (x_4 with a rescue rate of 79.23% and x_5 with 81.32%), x_3 shows higher

FIGURE 7 | Sensitivity analysis of parameter δ .

rescue efficiency under the same number of teams (39 teams), indicating more reasonable task assignment and route planning. Although x_1 and x_2 have slightly higher rescue rates (94.59% and 96.40%, respectively), they addressed fewer people in need (851 and 833 individuals). Moreover, they did not clearly reflect priority protection for low-connectivity areas. In contrast, x_3 incorporates the principle of “prioritizing vulnerable groups” into its dispatching logic, emphasizing the improvement of resource matching in low-connectivity areas. This better aligns with the requirements of fairness and humanitarian consideration in rescue operations.

6.2.2 | Sensitivity Analysis of δ

The parameter δ determines the uncertainty range of rescue demand at affected sites. This section performs a sensitivity analysis on δ , focusing on the performance evolution of five alternatives.

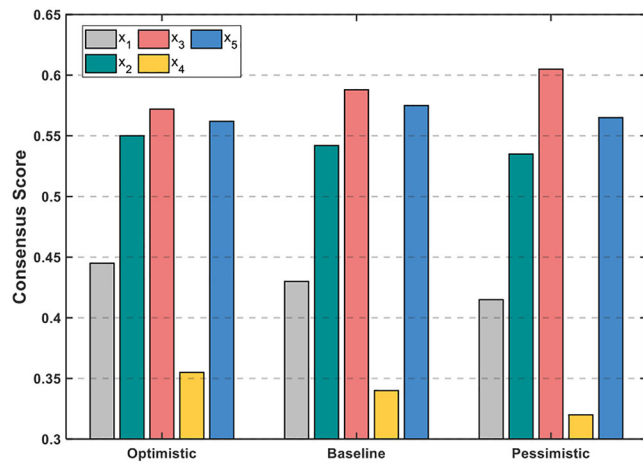
First, we examine the impact of the value of δ on decision results. Since this parameter directly governs the uncertainty of rescue demand, varying its value reflects different uncertainty scenarios. In this study, the value of δ is adjusted within the interval $[0.5\delta, 1.5\delta]$ to represent the uncertainty level. This variation covers scenarios ranging from relatively precise estimations to extreme ambiguity. Figure 7 illustrates the performance regarding the total number of rescued individuals for each alternative across these different values of δ .

As shown in Figure 7, as the value of δ increases to reflect a higher uncertainty level, the number of rescued individuals for all alternatives shows a downward trend. This reflects the objective principle that greater uncertainty degrades the efficiency of rescue scheduling. However, the performance ranking of the alternatives exhibits high stability. Regardless of the value of δ , x_3 consistently ranks first in terms of the number of rescued individuals. This indicates that x_3 can maintain optimal response efficacy even when the magnitude of error fluctuates drastically.

This section further examines the impact of the probability distribution of parameter δ on the optimal consensus support.

TABLE 10 | Dispatch plan metrics.

	x_1	x_2	x_3	x_4	x_5
Number of dispatched teams	39	39	39	39	39
Total rescued individuals	805	803	960	885	914
Individuals awaiting rescue	851	833	1117	1117	1124
Rescue rate	94.59%	96.40%	85.94%	79.23%	81.32%
Time window violation	394.12	342.50	406.14	392.39	456.10
Demand violation	7	2	66	88	89
Resource matching degree for low-connectivity areas	1.0216	0.9992	1.0398	0.9735	1.0587

**FIGURE 8** | Sensitivity Analysis of the probability distribution of parameter δ .

By adjusting the probability density of δ , we designed “Optimistic” and “Pessimistic” scenarios to compare with the baseline distribution. The Optimistic scenario simulates a state where rescue demand predictions are relatively precise by increasing the probability of low δ values; the Pessimistic scenario simulates a state where the predicted disaster situation is highly ambiguous by increasing the occurrence probability of high δ values. The results are shown in Figure 8.

Figure 8 displays the optimal consensus support for five alternatives under different probability distribution contexts. The results indicate that regardless of variations in the probability distribution, the ranking of alternatives consistently remains $x_3 > x_5 > x_2 > x_1 > x_4$. Notably, in the Pessimistic scenario, the advantage of alternative x_3 becomes even more pronounced. This reflects that x_3 holds higher decision-making value and has achieved a consistent consensus within the expert group.

6.2.3 | Stress Test Based on Robust Optimization (RO)

In decision theory under uncertainty, RO can be regarded as an extreme form of Distributionally RO (DRO) because it does not rely on specific probability distribution information but instead targets the worst-case realization within the parameter set. To validate the reliability of the proposed model under extreme

conditions, this section implements a stress test based on RO. We set the actual number of people to be rescued at all affected sites to their theoretical Upper Bound, simulating an extreme scenario where the disaster situation is completely out of control. Subsequently, we compare three decision models representing different methods of handling uncertainty: The Deterministic Model (DM), the SP, and the RO Model (RO). The performance of the optimal decision solutions for each model under this extreme scenario is presented in Table 11.

First, Table 11 reveals the vulnerability of the DM in extreme conditions. Since this model completely ignores fluctuations in the number of people awaiting rescue, its decision solutions lack necessary risk buffers. This causes the time window violation to surge to 890, and the actual number of rescued individuals is far lower than other models. This proves that in a highly uncertain disaster, an aggressive strategy that completely neglects risk is not only difficult to execute but may even lead to the collapse of the rescue network.

Second, by optimizing against the worst-case scenario, the RO model minimizes the risk of time window violations, demonstrating extremely high safety. However, this safety comes at the cost of overall rescue efficiency and fairness. To ensure absolutely no time window violations under extreme demand, the RO model adopts an overly conservative scheduling strategy. It proactively abandons a large number of affected areas that are distant or have high uncertainty, resulting in a total number of rescued people that is actually lower than that of the SP model.

Notably, the metric “Resource Matching Degree for Low-Connectivity Areas” reveals a potential ethical dilemma of conservative algorithms in humanitarian applications. Under the RO method, this metric scores only 0.8824, while the SP model maintains a higher level of 0.9835. This comparison indicates that to avoid systemic extreme risks, the RO algorithm tends to sacrifice vulnerable group areas that are remote, have weak information feedback capabilities, and have high demand uncertainty.

In summary, the stress test demonstrates that although the RO method theoretically provides defense against distributional deviations, its inherent “excessive conservativeness” conflicts with the first principle of “supremacy of life” in humanitarian rescue. In contrast, the SP framework proposed in this paper strikes a balance between efficiency and social fairness. Under

TABLE 11 | Performance comparison of models under stress test.

Evaluation metrics	DM	SP	RO
Total rescued individuals	750	882	855
Time window violation	890	520	210
Resource matching degree for lowConnectivity areas	0.7817	0.9835	0.8824

TABLE 12 | Comparison of existing GDM models.

Methods	AI decision	Alternatives	Methods of consensus-reaching	Application background
Z. Wu and Xu (2018)	No	Fixed	Aggregating similarity matrix	Emergency decision making
Xu et al. (2019)	No	Fixed	Aggregating similarity matrix	Emergency decision making
Tang et al. (2021)	No	Fixed	Distance of preferred relationships	Metro route building
Tong and Zhu (2023)	No	Fixed	Distance of preferred relationships	Product designing in industrial internet
J. Zhang et al. (2025)	Yes	Fixed	Distance of preferred relationships	Medical diagnosis
Our method	Yes	Dynamic	Maximum consensus sequence	Emergency decision making

the premise of assuming moderate and controllable risks, it upholds the principle of compassion, ensures that resources are directed towards vulnerable groups in low-connectivity areas, and maximizes the inclusiveness of the rescue operation.

6.3 | Comparison and Discussion

The theoretical and experimental results presented above demonstrate the effectiveness of the proposed method in disaster emergency decision-making. This section will emphasize the superiority of the proposed method by comparing it with existing research, as illustrated in Table 12.

1. Human–AI collaboration

As shown in Table 5, only our method and J. Zhang et al. (2025) focus on hybrid intelligence involving both humans and AI. J. Zhang et al. (2025) explored how to integrate human–AI interaction into the consensus reaching process and proposed a heterogeneous feedback strategy. In contrast, our method innovatively incorporates human–AI collaboration into the disaster emergency decision-making framework. Specifically, on the one hand, AI algorithms are used to estimate the needs of the affected population, addressing the limitations of human experts in handling large-scale data; on the other hand, in the final decision stage, the decision outcomes from both the crowd and AI are provided to the experts as feasible options, allowing them to perform comprehensive evaluations based on their experience and expertise. This human–AI collaborative approach not only leverages the strengths of AI in data processing and pattern recognition but also takes advantage of human experts' abilities in understanding complex issues and making value judgments,

thus enhancing the scientific rigor and effectiveness of the decision-making process.

2. Alternatives

In most existing studies (Z. Wu and Xu 2018; Xu et al. 2019; Tang et al. 2021; Tong and Zhu 2023; J. Zhang et al. 2025), the alternatives are usually predefined. Decision-makers select and evaluate from a fixed and limited set of options, which are typically determined through expert experience or historical data before the decision process begins and remain unchanged during the decision-making process. In contrast, our method generates the initial alternatives by combining a stochastic optimization model with the estimates from both the crowd and AI. Moreover, during the subsequent scenario generation and clustering processes, the alternatives are dynamically adjusted based on different clustering results. This dynamic generation and adjustment of alternatives allows for better adaptation to the complex disaster environment, reflecting the real-time needs of the affected population and the dynamic changes in rescue resources, providing decision-makers with more accurate and context-specific decision support.

3. Consensus-reaching methods

Existing studies have proposed two methods for calculating consensus measures. The first method integrates similarity matrices between each pair of subgroups to construct a consensus matrix, thereby determining the consensus level (Z. Wu and Xu. 2018; Xu et al. 2019). The second method combines the preferences of the subgroups to form a collective preference, which is then used as a reference point to calculate the consensus degree of each subgroup (Tang et al. 2021; Tong and Zhu 2023; J. Zhang et al. 2025). However, these methods depend on the choice of an integration function, and different functions may yield

varying results, which lack robustness. Furthermore, integration operations must be performed in each iteration, increasing computational complexity. In contrast, the method proposed in this paper is more straightforward, providing a direct ranking as the final consensus result, thereby enhancing the efficiency and timeliness of emergency rescue efforts.

4. Uncertainty handling

In traditional research on disaster emergency decision-making (Z. Wu and Xu 2018; Xu et al. 2019), it is typically assumed that the number of affected individuals and the demand for rescue resources are known and fixed. However, these assumptions are unrealistic in actual disaster scenarios. To address this, the paper introduces innovative methods for managing uncertainty. First, for the uncertainty of the number of affected people, a triangular probability distribution is used to generate multiple possible scenarios, and k -means clustering is applied to reduce the dimensionality of these scenarios, obtaining a representative set of scenarios. Then, in the stochastic optimization model, the probability distribution of these scenarios is fully considered, allowing the model to handle multiple sources of uncertainty simultaneously. This multidimensional approach to managing uncertainty offers a more comprehensive representation in the disaster decision-making process, enhancing the robustness and adaptability of the decisions.

7 | Conclusions, Implications and Future Directions

Crowdsourcing platforms and AI algorithms provide ways to ascertain the needs of disaster victims in real time. However, the shortcomings of both methods result in difficulties for humanitarian organizations that rely solely on either means for GDM. This situation inspired us to use crowd-AI intelligence for disaster emergency response. This study proposes a framework that can support humanitarian decision-making and obtain crowd-AI intelligence systematically. The applicability of the proposed framework is validated by a real-world case study of the 2021 Henan flood.

This study has the following theoretical and managerial implications.

Theoretical implications: This study innovatively proposes a combination of human and AI intelligence to address decision-making problems in humanitarian disaster response. Although previous studies have combined crowdsourcing techniques and AI algorithms, they have mainly concentrated on data processing tasks. There is a lack of research that integrates crowd and AI intelligence to address practical emergency response decision-making problems. This study also incorporates the real-time severity of the affected population's suffering into emergency response decision-making. In humanitarian operations, alleviating human suffering is the primary goal, as trapped individuals endure significant mental and physical stressors if deprived of essential goods and services. Our work is conducive to the real-time evaluation of human suffering, holding theoretical significance for integrating the subjective experiences of the affected population into top-level decision-making mechanisms.

Managerial Implications: For humanitarian decision-makers and crowdsourcing platform designers, this study offers valuable insights. Decision-makers can better understand how to take full advantage of crowd and AI intelligence and make corresponding decisions. By analyzing the shortcomings of these types of intelligence, decision-makers can strive to overcome them in practice. Furthermore, humanitarian decision-makers should emphasize the real-time suffering of the affected population, which plays a crucial role in the timely allocation of critical disaster relief resources.

Designers of disaster crowdsourcing platforms should be mindful of selection bias. They should create user-friendly interfaces that enable vulnerable groups, such as older individuals, to report information. Moreover, they should ensure the widespread dissemination of the platform to prevent people in remote areas from missing out on timely assistance.

However, our study has certain limitations. We use a text classification technique with the vulnerability feature system to determine urgency. Nevertheless, text and images conveyed by the affected population often express nuanced emotions. For instance, images depicting physical injuries may indicate physical pain, while images of destroyed homes or other distressing scenes could indicate mental stress. In the future, multimodal sentiment analysis should be used to discern sentiment intensity, particularly considering the emotional nuances contained in both text and images.

Acknowledgments

The work was supported by the National Natural Science Foundation of China (72401052, 72434001), the Doctoral Research Initiation Fund Program of Liaoning Province (2025-BS-0058), and the Fundamental Research Funds for the Central Universities (DUT23RC(3)052).

Conflicts of Interest

The authors declare no conflicts of interest.

References

- Abbasi, S., H. Vahdat-Nejad, and H. Moradi. 2024. "Harnessing Trustable Crowdsourcing Power for Flood Disaster Evaluation." *Natural Hazards* 120, no. 9: 8723–8741.
- Alhaffar, M. B. A., E. Joury, and A. Eriksson. 2023. "Community Engagement and Crowdsourcing for Effective Disaster Response and Rescue Operations During the Earthquake in Syria." *International Journal of Disaster Risk Reduction* 98: 104096.
- Alizadeh, B., D. Li, J. Hillin, et al. 2022. "Human-Centered Flood Mapping and Intelligent Routing Through Augmenting Flood Gauge Data With Crowdsourced Street Photos." *Advanced Engineering Informatics* 54: 101730.
- Bahadori, H., H. Vahdat-Nejad, and H. Moradi. 2022. "CrowdBIG: Crowd-Based System for Information Gathering From the Earthquake Environment." *Natural Hazards* 114, no. 3: 3719–3741.
- Chen, Y. L., and L. C. Cheng. 2009. "Mining Maximum Consensus Sequences From Group Ranking Data." *European Journal of Operational Research* 198, no. 1: 241–251.
- Cheng, C. S., A. Behzadan, and A. Noshadran. 2024. "A Post-Hurricane Building Debris Estimation Workflow Enabled by Uncertainty-Aware AI

- and Crowdsourcing." *International Journal of Disaster Risk Reduction* 112: 104785.
- Cheng, C. S., A. B. Khajwal, A. H. Behzadan, and A. Noshadravan. 2023. "A Probabilistic Crowd-AI Framework for Reducing Uncertainty in Postdisaster Building Damage Assessment." *Journal of Engineering Mechanics* 149, no. 9: 04023059.
- Cutter, S. L., J. T. Mitchell, and M. S. Scott. 2000. "Revealing the Vulnerability of People and Places: A Case Study of Georgetown County, South Carolina." *Annals of the Association of American Geographers* 90, no. 4: 713–737.
- Dargin, J. S., C. Fan, and A. Mostafavi. 2021. "Vulnerable Populations and Social Media Use in Disasters: Uncovering the Digital Divide in Three Major US Hurricanes." *International Journal of Disaster Risk Reduction* 54: 102043.
- Esparza, M., H. Farahmand, S. Brody, and A. Mostafavi. 2023. "Examining Data Imbalance in Crowdsourced Reports for Improving Flash Flood Situational Awareness." *International Journal of Disaster Risk Reduction* 95: 103825.
- Fallah, P., M. Rabiee, A. Yousefi-Babadi, E. Roghanian, and M. Hajiaghahi-Kesheteli. 2023. "Designing an Agile, Flexible and Resilient Disaster Supply Chain Network Using a Hybrid Group Decision-Making Robust Optimization Framework." *Computers & Industrial Engineering* 184: 109591.
- Fan, C., C. Zhang, A. Yahja, and A. Mostafavi. 2021. "Disaster City Digital Twin: A Vision for Integrating Artificial and Human Intelligence for Disaster Management." *International Journal of Information Management* 56: 102049.
- Fatemi, F., A. Ardalan, B. Aguirre, N. Mansouri, and I. Mohammadfam. 2017. "Social Vulnerability Indicators in Disasters: Findings From a Systematic Review." *International Journal of Disaster Risk Reduction* 22: 219–227.
- Flanagan, B. E., E. W. Gregory, E. J. Hallisey, J. L. Heitgerd, and B. Lewis. 2011. "A Social Vulnerability Index for Disaster Management." *Journal of Homeland Security and Emergency Management* 8, no. 1: 3.
- Ghaffarian, S., and S. Emtehani. 2021. "Monitoring Urban Deprived Areas With Remote Sensing and Machine Learning in Case of Disaster Recovery." *Climate* 9, no. 4: 58.
- Graves, J. M., D. A. Abshire, S. Amiri, and J. L. Mackelprang. 2021. "Disparities in Technology and Broadband Internet Access Across Ruralities: Implications for Health and Education." *Family & Community Health* 44, no. 4: 257–265.
- Guo, Q., S. Jiao, Y. Yang, Y. Yu, and Y. Pan. 2025. "Assessment of Urban Flood Disaster Responses and Causal Analysis at Different Temporal Scales Based on Social Media Data and Machine Learning Algorithms." *International Journal of Disaster Risk Reduction* 117: 105170.
- He, X., Y. Hu, X. Yang, S. Wang, and Y. Wang. 2024. "Urban Flood Resilience Evaluation Based on Heterogeneous Data and Group Decision-Making." *Entropy* 26, no. 9: 755.
- Holand, I. S., P. Lujala, and J. K. Rød. 2011. "Social Vulnerability Assessment for Norway: A Quantitative Approach." *Norsk Geografisk Tidsskrift-Norwegian Journal of Geography* 65, no. 1: 1–17.
- Holguín-Veras, J., M. Jaller, L. N. Van Wassenhove, N. Pérez, and T. Wachtendorf. 2012. "On the Unique Features of Post-disaster Humanitarian Logistics." *Journal of Operations Management* 30, no. 7–8: 494–506.
- Holguín-Veras, J., J. Amaya-Leal, V. Cantillo, L. N. Van Wassenhove, F. Aros-Vera, and M. Jaller. 2016. "Econometric Estimation of Deprivation Cost Functions: a Contingent Valuation Experiment." *Journal of Operations Management* 45: 44–56.
- Howe, J. 2006. "The Rise of Crowdsourcing." *Wired Magazine* 14, no. 6: 176–183.
- Huang, X., Z. Li, C. Wang, and H. Ning. 2020. "Identifying Disaster Related Social Media for Rapid Response: A Visual-Textual Fused CNN Architecture." *International Journal of Digital Earth*, 13, no. 9: 1017–1039.
- Kankanamge, N., T. Yigitcanlar, A. Goonetilleke, and M. Kamruzzaman. 2019. "Can Volunteer Crowdsourcing Reduce Disaster Risk? A Systematic Review of the Literature." *International Journal of Disaster Risk Reduction* 35: 101097.
- Keim, M. E., and E. Noji. 2011. "Emergent Use of Social media: a New Age of Opportunity for Disaster Resilience." *American Journal of Disaster Medicine* 6, no. 1: 47–54.
- Khajwal, A. B., and A. Noshadravan. 2024. "Enhancing Post-Hurricane Regional Loss Assessment With Crowdsourcing: A Probabilistic Inference Approach." *International Journal of Disaster Risk Reduction* 107: 104456.
- Lam, N. S., M. Meyer, M. Reams, et al. 2023. "Improving Social media Use for Disaster Resilience: Challenges and Strategies." *International Journal of Digital Earth* 16, no. 1: 3023–3044.
- Li, X. Y., and X. Wang. 2025. "Rescue Path Planning for Urban Flood: A Deep Reinforcement Learning-Based Approach." *Risk Analysis* 45, no. 4: 928–943.
- Liu, S., X. He, F. T. Chan, and Z. Wang. 2022. "An Extended Multi-Criteria Group Decision-Making Method With Psychological Factors and Bidirectional Influence Relation for Emergency Medical Supplier Selection." *Expert Systems with Applications* 202: 117414.
- Liu, Y., G. Wei, H. Liu, and L. Xu. 2022. "Group Decision Making for Internet Public Opinion Emergency Based Upon Linguistic Intuitionistic Fuzzy Information." *International Journal of Machine Learning and Cybernetics* 1–16.
- Loynes, C., J. Ouenniche, and J. De Smedt. 2022. "The Detection and Location Estimation of Disasters Using Twitter and the Identification of Non-Governmental Organisations Using Crowdsourcing." *Annals of Operations Research* 308, no. 1: 339–371.
- Martinez-Amaya, J., C. Radin, and V. Nieves. 2022. "Advanced Machine Learning Methods for Major Hurricane Forecasting." *Remote Sensing* 15, no. 1: 119.
- Moghadas, M., A. Fekete, A. Rajabifard, and T. Kötter. 2023. "The Wisdom of Crowds for Improved Disaster Resilience: A Near-Real-Time Analysis of Crowdsourced Social Media Data on the 2021 Flood in Germany." *GeoJournal* 88, no. 4: 4215–4241.
- Morte-Nadal, T., and M. Á Esteban-Navarro. 2025. "Recommendations for Digital Inclusion in the Use of European Digital Public Services." *Humanities and Social Sciences Communications* 12, no. 1: 1–14.
- Nielsen, A. B., D. Landwehr, J. Nicolai, T. Patil, and E. Raju. 2024. "Social media and Crowdsourcing in Disaster Risk Management: Trends, Gaps, and Insights From the Current State of Research." *Risk, Hazards & Crisis in Public Policy* 15, no. 2: 104–127.
- Ro, S. H., Y. Li, and J. Gong. 2024. "A Machine Learning Approach for Post-Disaster Data Curation." *Advanced Engineering Informatics* 60: 102427.
- Salmerón, J., and A. Apte. 2010. "Stochastic Optimization for Natural Disaster Asset Prepositioning." *Production and Operations Management* 19, no. 5: 561–574.
- Snezhana, D. 2023. "Applying Artificial Intelligence (AI) for Mitigation Climate Change Consequences of the Natural Disasters." *Research Journal of Ecology and Environmental Sciences* 3, no. 1: 1–8.
- Świtalski, S. 1999. "Control System With the Specific Performance Index for an Evaporator." *Control and Cybernetics* 28, no. 2: 315–329.
- Sun, W., P. Bocchini, and B. D. Davison. 2020. "Applications of Artificial Intelligence for Disaster Management." *Natural Hazards* 103, no. 3: 2631–2689.
- Tang, M., H. Liao, X. Mi, B. Lev, and W. Pedrycz. 2021. "A Hierarchical Consensus Reaching Process for Group Decision Making With Noncooperative Behaviors." *European Journal of Operational Research* 293, no. 2: 632–642.

- Tavra, M., I. Racetin, and J. Peroš. 2021. "The Role of Crowdsourcing and Social Media in Crisis Mapping: A Case Study of a Wildfire Reaching Croatian City of Split." *Geoenvironmental Disasters* 8: 1–16.
- Thekdi, S., U. Tatar, J. Santos, and S. Chatterjee. 2025. "On the Compatibility of Established Methods With Emerging Artificial Intelligence and Machine Learning Methods for Disaster Risk Analysis." *Risk Analysis* 45, no. 4: 863–877.
- Tong, H., and J. Zhu. 2023. "A Parallel Approach With the Strategy-Proof Mechanism for Large-Scale Group Decision Making: An Application in Industrial Internet." *European Journal of Operational Research* 311, no. 1: 173–195.
- Wang, H., J. L. Lin, Z. S. Chen, and Z. Wang. 2025. "Large-Scale Group Decision Making for Simulation-Guided Disaster Recovery Center Location Selection." *Information Sciences* 700: 121854.
- Wang, X., X. D. Liang, X. Y. Li, and P. Luo. 2022. "An Integrated Multiattribute Group Decision-Making Approach for Risk Assessment in Aviation Emergency Rescue." *International Journal of Aerospace Engineering* 2022, no. 1: 9713921.
- Wang, Y., B. Yan, W. Xia, X. Hu, H. Ma, and P. Jin. 2022. "Personalization-Driven Consensus Reaching Model for Emergency Mission Scheduling Schemes Selection in Large-Group Emergency Decision-Making With Linguistic Distribution Preference Relationship." *International Journal of Fuzzy Systems* 24, no. 7: 3308–3326.
- Wang, Z., N. S. Lam, M. Sun, et al. 2022. "A Machine Learning Approach for Detecting Rescue Requests From Social Media." *ISPRS International Journal of Geo-Information* 11, no. 11: 570.
- Wei, X., Z. Zhang, M. Y. Zhang, W. Y. Chen, and D. D. Zeng. 2022. "Combining Crowd and Machine Intelligence to Detect False News on Social Media." *MIS Quarterly* 46, no. 2: 977–1008.
- Wu, X., and H. Liao. 2024. "A Multi-Stage Multi-Criterion Group Decision-Making Method for Emergency Management Based on Alternative Chain and Trust Radius of Experts." *International Journal of Disaster Risk Reduction* 101: 104253.
- Wu, Z., and J. Xu. 2018. "A Consensus Model for Large-Scale Group Decision Making With Hesitant Fuzzy Information and Changeable Clusters." *Information Fusion* 41: 217–231.
- Xinhua News Agency. 2022, January 21. "Investigation Report on the '7·20' Extreme Rainstorm Disaster in Zhengzhou, Henan Released [EB/OL]." *The State Council of the People's Republic of China*. https://www.gov.cn/xinwen/2022-01/21/content_5669723.htm.
- Xu, X. H., Z. J. Du, X. H. Chen, and C. G. Cai. 2019. "Confidence Consensus-Based Model for Large-Scale Group Decision Making: A Novel Approach to Managing Non-Cooperative Behaviors." *Information Sciences* 477: 410–427.
- Yan, S., Y. Xu, Z. Gong, and E. Herrera-Viedma. 2025. "Fuzzy Quantum Group Decision Making and Its Application in Meteorological Disaster Emergency." *IEEE Transactions on Fuzzy Systems* 33, no. 5: 1441–1454.
- Yoo, C., E. Yoo, L. Yan, and A. Pedraza-Martinez. 2024. "Speak With One Voice? Examining Content Coordination and Social Media Engagement During Disasters." *Information Systems Research* 35, no. 2: 551–569.
- Zhang, H., X. Zhao, X. Fang, and B. Chen. 2024. "Proactive Resource Request for Disaster Response: A Deep Learning-Based Optimization Model." *Information Systems Research* 35, no. 2: 528–550.
- Zhang, J., N. Wang, and M. Tang. 2025. "Human-AI Coordination for Large-Scale Group Decision Making With Heterogeneous Feedback Strategies." *Journal of the Operational Research Society* 76: 1–21.
- Zhao, Y., and Q. Zhu. 2014. "Evaluation on Crowdsourcing Research: Current Status and Future Direction." *Information Systems Frontiers* 16: 417–434.
- Zhu, S., J. Huang, and Y. Xu. 2021. "A Consensus Model for Group Decision Making With Self-Confident Linguistic Preference Relations." *International Journal of Intelligent Systems* 36, no. 11: 6360–6386.

Appendix

TABLE A1 | Rescue teams and their staffing.

Rescue team	Maximum rescue capacity
T1	40
T2	38
T3	55
T4	26
T5	28
T6	18
T7	40
T8	29
T9	60
T10	32
T11	10
T12	27
T13	22
T14	16
T15	50
T16	40
T17	36
T18	24
T19	64
T20	68
T21	36
T22	95
T23	40
T24	50
T25	32
T26	30
T27	46
T28	50
T29	45
T30	30
T31	30
T32	48
T33	24
T34	12
T35	45
T36	20
T37	65
T38	40
T39	56
T40	20
T41	40
T42	82

Crowd Scenarios (First-Stage Rescue Team Dispatch Plan)

TABLE A2 | First-stage rescue team deployment arrangement.

Starting point	Dispatched team numbers	Number of dispatched teams
a20	1,2,10,13,14,17,19,20,22,28,30,31,32,35,39	15
a51	3,5,7,8,9,15,16,18,24,25,26,29,34,36,37,38,42	17
a135	4,6,11,23,33,40,41	7

Crowd Scenarios (Second-Stage Rescue Team Dispatch Plan)

Crowd Scenario 1

TABLE A3 | Detailed rescue route for Crowd Scenario 1.

Starting point	Rescue team	
a20	1	a20(0) → a 108 (1)→ a105 (6)→a 20
	2	a20 (5)→ a14 (3)→ a109(9) → a116 (16)→a 20
	10	a20(0) → a21(20)→ a23(1) → a6(2) →a20
	13	a20(0) → a109(11) →a 20
	14	a20(0) → a32(3) → a33(3) → a116(3) →a 20
	17	a20(0) → a11(3) → a2(2) → a20
	19	a20(0) → a102(2) → a113(4) → a20
	20	a20(0) → a25(3) → a100(43) → a20
	22	a20(0) → a30(22) → a114(6) → a20
	28	a20(5) → a116(16) →a 15(14) → a20
	30	a20(0) → a100(27) → a20
	31	a20(11) → a1(3) → a13(5) → a20
	32	a20(1) → a28(5) → a100(28) →a 107(2) → a20
	35	a20(0) → a118(11) → a12(6) → a20
	39	a20(0) → a131(7) → a57(8) → a20
a51	4	a51(0) → a41(2) → a51
	6	a51(0) → a106(6) → a16(2) → a5(7) →a 51
	11	a51(0)→ a110(1) → a14(3) → a51
	23	a51(0) → a125(13) → a112(20) → a51
	33	a51(0) → a112(13) → a15(2) → a51
	40	a51(0) → a27(5) → a51
	41	a51(0) → a8(3) → a29(9) → a4(2) →a 51
a135	3	a135(0) → a112(8) → a10(8) → a135
	5	a135(0) → a16(7) →a 114(16) →a 135
	7	a135(3) →a 25(17) → a120(2) → a56(10) →a135
	8	a135(0) → a125(3) → a112(18) → a135
	9	a135(0) → a30(35) → a13(3) → a110(2) → a135
	15	a135(0) → a32(12) → a17(21) → a135

(Continues)

TABLE A3 | (Continued)

Starting point	Rescue team
16	a135(0) → a100(25) → a32(5) → a135
18	a135(0) → a28(3) → a135
24	a135(3) → a30(21) → a31(8) → a135
25	a135(6) → a23(4) → a20(9) → a24(3) → a135
26	a135(0) → a108(2) → a125(7) → a100(17) → a135
29	a135(0) → a21(28) → a135
34	a135(0) → a23(8) → a124(3) → a135
36	a135(0) → a117(3) → a24(7) → a135
37	a135(0) → a18(20) → a26(3) → a17(14) → a135
38	a135(6) → a131(6) → a100(17) → a135
42	a135(0) → a31(10) → a135

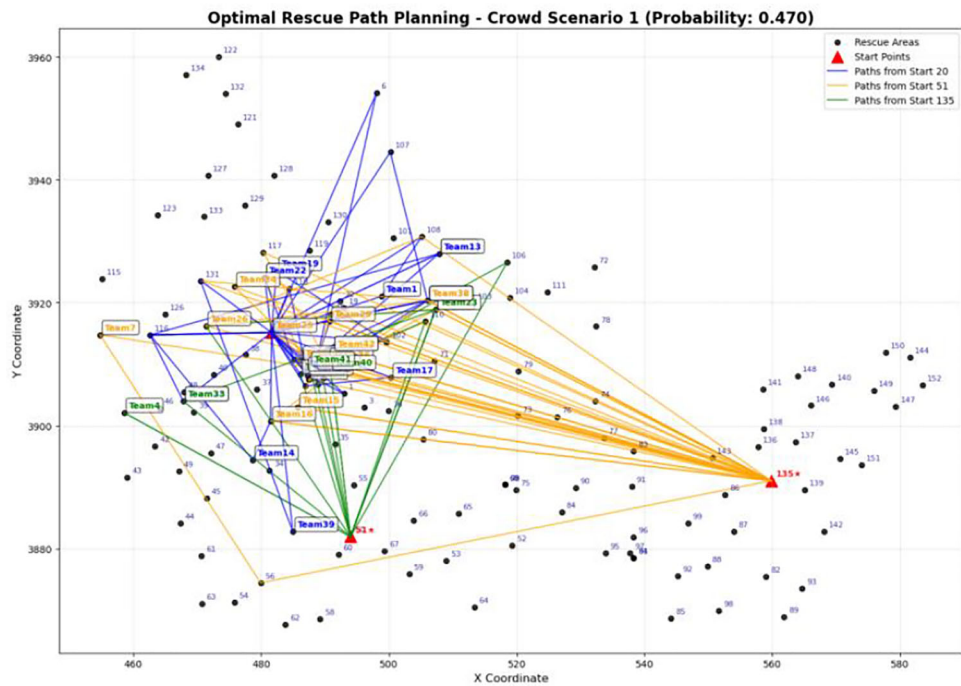


FIGURE A1 | Rescue team dispatch route for Crowd Scenario 1.

TABLE A4 | Detailed rescue route for Crowd Scenario 2.

Starting point	Rescue team	Rescue path
a20	1	a20(0) → a107(3) → a112(31) → a20
	2	a20(0) → a16(6) → a109(12) → a32(9) → a20
	10	a20(0) → a100(25) → a20
	13	a20(0) → a30(7) → a31(6) → a20
	14	a20(0) → a63(1) → a102(2) → a20
	17	a20(0) → a23(10) → a114(17) → a20
	19	a20(0) → a110(4) → a41(2) → a105(6) → a20
	20	a20(18) → a30(34) → a20
	22	a20(18) → a68(2) → a116(27) → a20
	28	a20(0) → a18(18) → a15(15) → a108(4) → a20
	30	a20(0) → a112(24) → a20
	31	a20(0) → a131(14) → a6(4) → a20
	32	a20(0) → a2(1) → a100(31) → a20
	35	a20(0) → a28(4) → a16(1) → a30(22) → a20
	39	a20(0) → a109(7) → a20
a51	4	a51(0) → a21(20) → a26(2) → a51
	6	a51(0) → a70(1) → a17(11) → a51
	11	a51(0) → a30(5) → a112(0) → a33(1) → a51
	23	a51(0) → a25(23) → a5(7) → a51
	33	a51(0) → a120(2) → a29(9) → a51
	40	a51(0) → a100(13) → a23(1) → a51
	41	a51(0) → a100(27) → a30(3) → a51
a135	3	a135(0) → a11(3) → a27(7) → a12(6) → a135
	5	a135(6) → a21(12) → a118(6) → a135
	7	a135(7) → a125(25) → a135
	8	a135(0) → a10(7) → a31(11) → a135
	9	a135(0) → a17(29) → a135
	15	a135(0) → a24(10) → a1(6) → a114(4) → a135
	16	a135(0) → a32(5) → a13(10) → a135
	18	a135(0) → a57(8) → a135
	24	a135(0) → a100(35) → a106(4) → a135
	25	a135(0) → a118(7) → a113(4) → a135
	26	a135(0) → a124(3) → a117(3) → a135
	29	a135(0) → a21(12) → a8(3) → a135
	34	a135(0) → a30(7) → a135
	36	a135(0) → a4(2) → a106(1) → a135
	37	a135(0) → a56(12) → a135
	38	a135(6) → a116(12) → a32(6) → a135
	42	a135(0) → a14(6) → a103(1) → a28(4) → a135

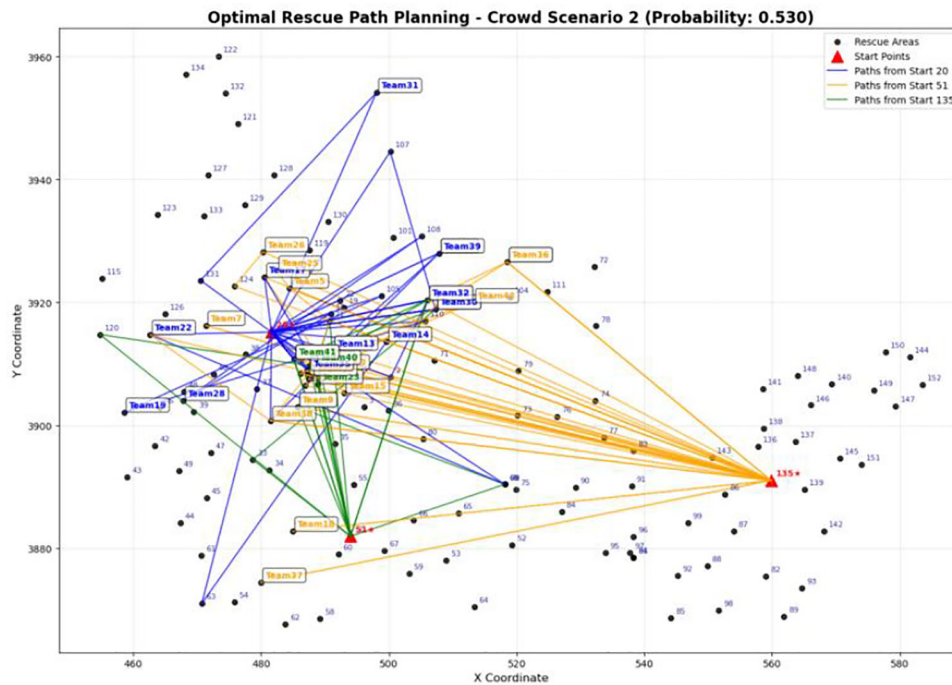


FIGURE A2 | Rescue team dispatch route for Crowd Scenario 2.

Machine Scenarios (First-Stage Rescue Team Dispatch Plan)

TABLE A5 | First-stage rescue team deployment arrangement.

Starting point	Dispatched team numbers	Number of dispatched teams
a20	2,6,13,15,16,18,21,22,24,25,26,32,35,41	14
a51	8,10,11,12,14,17,19,27,30,31,33,40	12
a135	1,3,7,20,23,28,29,34,36,37,38,39,42	13

TABLE A6 | Detailed rescue route for Machine Scenario 1.

Starting point	Rescue team	Rescue path
a20	2	a20(0) → a56(12) → a12(9) → a20
	6	a20(0) → a1(6) → a20
	13	a20(1) → a31(15) → a19(1) → a20
	15	a20(0) → a27(9) → a114(2) → a20
	16	a20(2) → a8(2) → a118(25) → a20
	18	a20(9) → a135(9) → a3(1) → a20
	21	a20(0) → a25(23) → a20
	22	a20(24) → a81(2) → a113(46) → a20
	24	a20(0) → a136(3) → a10(13) → a20
	25	a20(0) → a108(3) → a32(20) → a20
	26	a20(0) → a16(11) → a29(13) → a20
	32	a20(0) → a100(26) → a50(1) → a41(11) → a20
	35	a20(7) → a30(22) → a20
	41	a20(0) → a24(10) → a125(15) → a20
a51	8	a51(6) → a57(8) → a30(11) → a51
	10	a51(0) → a30(10) → a105(11) → a51
	11	a51(0) → a103(1) → a1(3) → a28(3) → a51
	12	a51(0) → a106(7) → a40(1) → a26(6) → a51
	14	a51(0) → a35(2) → a120(8) → a51
	17	a51(0) → a116(19) → a4(10) → a51
	19	a51(0) → a14(14) → a18(32) → a51
	27	a51(0) → a13(7) → a34(9) → a51
	30	a51(8) → a109(15) → a51
	31	a51(0) → a68(20) → a51
	33	a51(0) → a114(16) → a2(1) → a51
	40	a51(0) → a57(12) → a51
a135	1	a135(0) → a5(9) → a33(8) → a135
	3	a135(0) → a68(37) → a39(7) → a135
	7	a135(18) → a117(1) → a23(7) → a135
	20	a135(0) → a23(8) → a135
	23	a135(14) → a105(6) → a32(11) → a135
	28	a135(0) → a11(2) → a17(31) → a135
	29	a135(21) → a113(19) → a135
	34	a135(0) → a118(5) → a112(5) → a135
	36	a135(0) → a31(14) → a117(1) → a135
	37	a135(3) → a21(28) → a135
	38	a135(16) → a31(16) → a135
Machine Scenario 1	39	a135(0) → a15(11) → a131(13) → a135
	42	a135(0) → a100(60) → a7(1) → a135

TABLE A7 | Detailed rescue route for Machine Scenario 2.

Starting point	Rescue team	Rescue path
a20	2	a20(1) → a31(18) → a23(9) → a20
	6	a20(0) → a109(6) → a39(4) → a12(3) → a20
	13	a20(0) → a105(17) → a20
	15	a20(0) → a16(12) → a20
	16	a20(0) → a24(9) → a20
	18	a20(0) → a14(13) → a103(1) → a20
	21	a20(0) → a4(11) → a20
	22	a20(0) → a30(26) → a13(7) → a20
	24	a20(0) → a112(20) → a5(6) → a114(14) → a20
	25	a20(0) → a23(6) → a56(2) → a109(7) → a20
	26	a20(0) → a113(2) → a30(15) → a112(4) → a20
	32	a20(5) → a31(19) → a135(15) → a20
	35	a20(22) → a21(18) → a20
	41	a20(3) → a17(13) → a112(16) → a20
	41	a20(3) → a17(13) → a112(16) → a20
a51	8	a51(8) → a113(14) → a51
	10	a51(0) → a29(11) → a17(15) → a51
	11	a51(0) → a24(3) → a118(3) → a51
	12	a51(0) → a41(11) → a51
	14	a51(3) → a20(1) → a125(5) → a51
	17	a51(0) → a9(8) → a51
	19	a51(0) → a25(13) → a28(11) → a51
	27	a51(0) → a21(10) → a51
	30	a51(0) → a100(20) → a116(1) → a51
	31	a51(0) → a116(6) → a68(12) → a51
	33	a51(5) → a10(13) → a51
	40	a51(0) → a18(4) → a32(11) → a51
a135	1	a135(4) → a18(21) → a56(7) → a135
	3	a135(0) → a116(7) → a120(9) → a113(27) → a135
	7	a135(5) → a113(20) → a15(5) → a135
	20	a135(0) → a34(8) → a118(34) → a135
	23	a135(12) → a25(14) → a135
	28	a135(0) → a68(29) → a1(8) → a135
	29	a135(18) → a112(18) → a135
	34	a135(0) → a100(4) → a109(3) → a33(1) → a135
	36	a135(5) → a125(5) → a116(5) → a135
	37	a135(1) → a106(6) → a31(13) → a135
	38	a135(13) → a33(3) → a20(13) → a135
	39	a135(0) → a27(5) → a100(34) → a135
	42	a135(0) → a12(5) → a131(12) → a135

TABLE A8 | Detailed rescue route for Machine Scenario 3.

Starting point	Rescue team	Rescue path
a20	2	a20(0) → a13(7) → a57(8) → a16(6) → a20
	6	a20(0) → a17(9) → a137(3) → a20
	13	a20(0) → a15(9) → a116(9) → a20
	15	a20(0) → a31(11) → a68(28) → a7(1) → a20
	16	a20(0) → a10(5) → a113(23) → a114(1) → a20
	18	a20(4) → a8(1) → a105(6) → a116(6) → a20
	21	a20(0) → a68(19) → a20
	22	a20(0) → a25(7) → a118(30) → a20
	24	a20(15) → a17(10) → a30(13) → a20
	25	a20(0) → a18(29) → a20
	26	a20(0) → a40(1) → a106(6) → a20
	32	a20(0) → a114(7) → a30(15) → a28(8) → a20
	35	a20(6) → a21(14) → a109(13) → a20
	41	a20(0) → a39(10) → a27(9) → a20
a51	8	a51(0) → a10(5) → a30(10) → a118(10) → a51
	10	a51(0) → a32(13) → a100(14) → a51
	11	a51(0) → a105(8) → a70(1) → a51
	12	a51(0) → a5(8) → a51
	14	a51(3) → a12(1) → a114(3) → a116(3) → a51
	17	a51(3) → a100(26) → a51
	19	a51(0) → a41(12) → a131(13) → a51
	27	a51(0) → a23(14) → a102(2) → a125(19) → a51
	30	a51(0) → a81(2) → a20(23) → a51
	31	a51(7) → a100(18) → a51
	33	a51(0) → a113(11) → a16(6) → a51
	40	a51(0) → a9(5) → a17(12) → a51
a135	1	a135(0) → a68(16) → a114(6) → a14(2) → a135
	3	a135(12) → a21(13) → a33(4) → a25(6) → a135
	7	a135(0) → a31(35) → a135
	20	a135(0) → a35(2) → a113(28) → a135
	23	a135(0) → a56(10) → a124(3) → a135
	28	a135(28) → a14(8) → a135
	29	a135(22) → a34(4) → a25(13) → a135
	34	a135(0) → a50(1) → a100(9) → a135
	36	a135(0) → a57(12) → a135
	37	a135(0) → a31(7) → a135
	38	a135(14) → a8(1) → a1(11) → a135
	39	a135(0) → a33(4) → a29(13) → a135
	42	a135(0) → a107(4) → a4(11) → a24(13) → a135

